Getting to know Dell PowerFlex v3.6.x

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Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Getting to Know PowerFlex

The following section provides an overview of PowerFlex.

1

Overview of PowerFlex

PowerFlex is an enterprise-class, software-defined block storage solution that is deployed, managed, and supported as a single system. Highly flexible and scalable, PowerFlex allows customers to mix two-layer (Server SAN), single-layer HCI, storage-only, and other mixed architectures within a single deployment.

With PowerFlex, resources such as storage and compute can be scaled together or separately, non-disruptively, and in small increments. The system can scale from just a few nodes to hundreds in a cluster, linearly scaling I/O performance and throughput. In addition, PowerFlex supports a broad set of operating environments that include multiple hypervisors, bare-metal operating systems, and container management tools. This allows customers to flexibly transition application architectures, while supporting multiple generations of applications with disparate architectures in a single system.

PowerFlex offers integrated product options. PowerFlex appliance is a scalable system with flexible form factors that comes pre-configured and validated for fast, easy deployment. PowerFlex rack is a rack-scale system that is manufactured, managed, supported, and sustained as one system for single-end-to-end lifecycle support.

PowerFlex Manager provides IT operations management for Dell PowerFlex systems. It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. PowerFlex Manager can also be used to deploy and manage new and existing PowerFlex rack and PowerFlex appliance systems.

New features for PowerFlex

Learn about the new features introduced in PowerFlex 3.6-<build> software.

- PowerFlex is now available as a consumption model in the public cloud. PowerFlex 3.6.0.4 is supported on .
- PowerPath 7.4 for Linux now supports PowerFlex devices. PowerPath Migration enabler can be used for data migration across PowerFlex and other PowerPath supported storage array classes.

  NOTE: Only PowerFlex SCSI devices are supported. Since PowerFlex devices have only a single path, PowerFlex takes care of load balancing across different SDS.

Important

The following aspects of PowerFlex 3.6-<build> must be understood before deploying or upgrading your system:

1. CloudIQ integration
   - Discovering a PowerFlex system in CloudIQ requires the system to be registered with Secure remote services. In the PowerFlex Installer, **Maintain** tab, ensure that the PowerFlex system is registered with Secure remote services and that **Send data to CloudIQ** is enabled.
   - After installing a *new* PowerFlex system with PowerFlex version 3.6, it takes 24 hours until data is available in CloudIQ. Systems *upgrading* to PowerFlex 3.6 will appear in CloudIQ within an hour of enabling the **Send data to CloudIQ** option.
   - PowerFlex appliance and rack systems need PowerFlex Manager version "H" in order to be discovered in CloudIQ.
2. Replication enhancements
   - Bi-directional replication between two sites only is supported.
   - See the PowerFlex Sizer for sizing guidelines for correct replication system specifications, including WAN bandwidth sizing. When using the PowerFlex Sizer, take into account existing workload and existing hardware, and the possible need to add additional nodes to enable replication without impacting the current workload.
   - Replication capability has higher I/O overhead, as explained in the v3.6 *Getting to Know PowerFlex Guide* (Replication section) and the white paper Dell EMC PowerFlex: Introduction to Replication. See these recommendations in order to size your system correctly, and ensure that you have sufficient existing storage capabilities to enable this ability.
3. Network resiliency improvements
   - Proactive network path error detection requires SDC to be at v3.6 or later.
4. Upgrade Paths
   - Upgrade from PowerFlex version 3.0.x and 3.5.x to version 3.6 are supported.
   - Older PowerFlex/ VxFlex operating system versions must be upgraded to version 3.0.x prior to upgrading to version 3.6.

5. vSphere plug-in End-of-Support-Life
   - The PowerFlex vSphere plug-in continues to be supported with vSphere versions 6.5 and 6.7, however it reaches end-of-support-life when vSphere 6.5 and 6.7 reach end-of-support by VMware. For details on the last release of vSphere plug-in due to end of general support for ESXi 6.5 and 6.7 by VMware, see: Announcing End of General Support for vSphere 6.5.x and vSphere 6.7.x (89305). Also note the new issue in the *Known Issues* section that is related to vSphere Plug-in.
   - The PowerFlex vSphere plug-in does not support vSphere 7 hyperconverged (HCI) configurations.

Workflow of PowerFlex

The following steps describe the order to follow to get the PowerFlex system up and running in your environment:

1. Assemble your hardware according to your requirements for your environment.
2. Connect and configure the network.
3. Install the Operating System and prerequisites for PowerFlex.
4. Deploy PowerFlex. Refer to the *Deployment Guide* for step by step deployment procedures.
5. Allocate volumes and map them to hosts.
6. Begin using the PowerFlex storage.

2

Supported Product Limits and System Requirements

The following topics describe the supported product limits and system requirements for the PowerFlex system.

Product limits

The following table summarizes PowerFlex supported capabilities and system limits. The PowerFlex GUI is not backward compatible.

NOTE: The supported limits described might differ from system enforced limits in the list output of the `scli --query_system_limits` command.

Table 1. Supported capabilities and system limits

PowerFlex Item	Limit
System raw capacity	Up to 16 PB **NOTE:** In earlier versions of PowerFlex, there was an issue with scale limits. Due to certain scale limits, customers who had systems with an allocated capacity higher than 9 PB needed to consult with PowerFlex product experts for good configurations. This issue is fixed in PowerFlex v4.0 and PowerFlex v3.6.1.
Device size	Minimum: 240 GB, Maximum: 8 TB
Minimum number of SDSs in a Storage Pool	3
Minimum Storage Pool size	720 GB
Minimum devices (drives) per Storage Pool	3, one per Fault Unit
Volume size	Minimum: 8 GB, Maximum: 1 PB
Maximum filesystem partitions per volume	15
Maximum total number of volumes and snapshots in system	131,072 [a]
Maximum total number of volumes and snapshots in Protection Domain	32,768
Maximum total number of volumes and snapshots per Storage Pool	32,768
Maximum number of snapshots in a single V-Tree	126
Maximum raw capacity per Protection Domain	8 PB
Maximum raw capacity per SDS	128 TB
Maximum SDCs per system	2048
Maximum SDSs per system	512 [a]
SDSs per Protection Domain	128 [a]
Maximum devices (drives) per SDS server	64 [b]

Table 1. Supported capabilities and system limits (continued)

PowerFlex Item	Limit
NOTE: Includes NVDIMM and/or RFcache devices.	
Maximum devices per Protection Domain	8192
Maximum devices per Storage Pool	300 [a]
Maximum length of the SDS storage device path	63
Maximum acceleration devices of type RFcache (only one Acceleration Pool supported)	8 [c], [d]
Total size of all volumes per Storage Pool	4 PB
Maximum acceleration devices of type NVDIMM (maximum supported in node)	6 [c], [d]
Maximum volumes that are mapped to a single SDC	• ESXi 6.0 256-volumes • ESXi 6.5 512-volumes • ESXi 6.7, 7.0 1024-volumes Other operating systems: 1024
System Over Provisioning Factor	5X of net capacity per MG layout
Fine-Granularity maximum compression	10X of raw capacity
Maximum Protection Domains per system	256
Maximum Storage Pools per system	1024
Maximum Storage Pools per Protection Domain	64
Maximum Fault Sets per Protection Domain	64 [a]
RMcache	Minimum: 128 MB, Maximum: 300 GB
RFcache	• The RFcache feature is only supported by 512 Byte devices for the cache use case. RFcache is not supported on 4K devices used for cache. However, RFcache supports acceleration of 4K storage devices. • RFcache capability support is available only on Linux Kernel versions lower than 5.x
Maximum number of snapshots a Snapshot Policy can be defined to retain per V-Tree (not including locked snapshots)	60
Maximum Snapshot Policies per system	1000
Maximum user accounts	256 [a]
Maximum number of concurrent logged-in management clients (GUI/REST/CLI)	128
Maximum volumes that can be mapped via API (scli \REST\WebUI) concurrently	1024
Maximum number of configured syslog servers	16
Volumes per local Consistency Group (snapshot)	1024
Maximum number of volume to SDC mappings per system	262,143
SCSI-3 reservation type	Write exclusive registrants only
Replication Limits	
Number of destination systems for replication	1

Table 1. Supported capabilities and system limits (continued)

PowerFlex Item	Limit
Maximum number of SDR per system	128
Maximum number of Replication Consistency Group (RCG)	1024
Maximum replication pairs in RCG with initial copy	1024
Maximum number of Volume Pairs per RCG	1024
Maximum Volume Pairs per system	32,000
Maximum number of remote Protection Domains	8
Maximum number of copies per RCG	1
Maximum ratio of peer SDRs in a source protection domain to a target protection domain	1:3
Maximum ratio of maintenance SDRs in a source protection domain to a target protection domain	1:6
Recovery Point Objective (RPO)	Minimum: 15 seconds, Maximum: 1 hour
Maximum replicated volume size	64 TB

a. For further support ,contact Customer Support.
b. On a VMware server, the maximum devices per SDS is 59.
c. An Acceleration Pool can serve multiple Storage Pools, but a single Storage Pool can't use multiple Acceleration Pools.
d. An acceleration device may serve multiple SDS devices, but SDS devices can't span across multiple acceleration devices.

Table 2. Deployment and NDU Limits

PowerFlex Item	Limit
Maximum latency for MDM network	200 msec
SDS IP addresses	Must be unique. For example, 127.0.0.1 must not be used, as it refers to several machines
vSphere PowerFlex plug-in	• Does not support unmapping a volume from the SDC if it's disconnected • Does not support VMDK and RDM mixed environments • Adobe flash player is EOL as of 2020-12-31. To continue using the vSphere PowerFlex plug-in, follow the workaround described in the following article: https://www.dell.com/support/kbdoc/en-us/000181275/scaleio-vxflex-adobe-flash-end-of-life-powerflex-vsphere-plugin-workaround
RFcache capability support is available only on Linux Kernel versions lower than 5.X.	For example, supported RHEL versions: • RHEL 8.X Kernel Base: 4.18.0 • RHEL 7.X Kernel Base: 3.10.0

System requirements

This section lists the requirements for system components.

This section is specific to PowerFlex software deployments.

PowerFlex cluster components

Each PowerFlex cluster includes the following components. Before deploying PowerFlex, ensure that the PowerFlex environment meets the server requirements for the cluster type.

Three-node cluster

- One primary MDM
- One secondary MDM
- One tiebreaker
- Minimum of three SDSs (on the same servers as the above components, or on three different servers)
- SDCs, up to the maximum allowed (on the same servers as the above components, or on different servers)

Five-node cluster

- One primary MDM
- Two secondary MDMs
- Two tiebreakers
- Minimum of three SDSs (on the same servers as the above components, or on three different servers)
- SDCs, up to the maximum allowed (on the same servers as the above components, or on different servers)

For additional information, see the Getting to Know Dell PowerFlex Guide.

Additional requirements

PowerFlex Gateway server on a separate server, or together with an SDS. Do not install the PowerFlex Gateway on an SDC server.

NOTE: This limitation is required as to avoid cases when doing NDU and the SDC upgrade requires a reboot. This will cause the NDU to stop since the PowerFlex Gateway is managing it. If you are using the PowerFlex Gateway as a REST gateway, it is recommended to be on a separate server, as doing maintenance to servers with other storage roles enables the REST gateway to continue to function.

Physical server requirements

The following table summarizes the requirements of the physical server.

PowerFlex offers a sizing tool to make the complex work of sizing the physical requirement. Go to https://scaleio-sizer.emc.com/sizer and review the physical sizing considerations on the page. It is recommended to use the sizer tool for all and not perform manual sizing.

Table 3. Physical server requirements

Component	Requirement
Processor	One of the following: • Intel or AMD x86 64-bit (recommended) • Intel or AMD x86 32-bit (for Citrix Hypervisor (Xen) only) **NOTE:** AMD processors are not supported on VxFlex Ready Node and do not support NVDIMMs. For this reason, there is no support for Fine Granularity Storage Pools on AMD based servers. CPU threads: • MDM: 5 • SDS: Default 8/Max 12 • CloudLink: 2 • LIA: 1 • SDR: 8

Table 3. Physical server requirements (continued)

<table>
<tr><th>Component</th><th>Requirement</th></tr>
<tr><td></td><td>
SDC: Default 2/Max 8
NOTE: The recommended best practice is to have a different NUMA domain then SDR if it's on the same OS.
</td></tr>
<tr><td>Physical memory</td><td>
PowerFlex component requirements:
RAM (GiB)

SDS FG : 10 GiB + ((100*Number_Of_Drives) + (550 * Total_Drive_Capacity_in_TiB))/1024
SDS MG : 5 GiB + (210*Total_Drive_Capacity_in_TiB)/1024
SDS NVDIMM for FG: NVDIMM_capacity_in_GiB=((100*Number_Of_Drives) + (700 * Total_Drive_Capacity_in_TiB))/1024
MDM : 6.5 GiB
Cloudlink : 4 GiB
Operating System : 1 GiB
LIA : 0.35 GiB
PowerFlex Installer/PowerFlex Gateway: 8 GiB
VASA: Default 8GiB/Max 16 GiB
SDC: 6 GiB
SDR : 8586 MiB + 550 MiB * (max remote PDs count)

NOTE: For certain system configurations, the SDS needs to be configured to be able to use more than one Non-Uniform Memory Access (NUMA) domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and therefore, by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0. If NUMA 0 (usually half the total memory) is less than the memory required by the SDS, you need to allow the SDS access to the memory in the other NUMA. For more information, see the explanation and configuration procedures for "Configure the SDS for optimal memory usage" in "Post-deployment tasks" in the Deploy Dell PowerFlex Guide.
Non-Volatile RAM (NVDIMM):

Node Storage Capacity < 49 TiB : NVDIMM size 32 GiB
Node Storage Capacity between [49 TiB, 96 TiB] : NVDIMM size 64 GiB
Node Storage Capacity > 96 TiB : NVDIMM size 96 GiB

NOTE: PowerFlex does not support using SWAP on Storage only and SVM nodes. From version 3.0 and above, this should be the default setting:

Using SWAP with PowerFlex has the following impact:

Slows the performance of the system
Impacts I/O on the system disks which is used for SWAP; this may cause other functions using the system disk such as to have slow I/Os:

MDM repository related operations
PowerFlex logs written to disk. Refer to the following link on how to disable SWAP: https://www.tecmint.com/disable-swap-partition-in-centos-ubuntu/. In addition, the settings for vm.overcommit must be set to the following settings on nodes that have MDM and SDS process on them: <code>in/etc/sysctl.conf</code> file
<pre>vm.overcommit_memory=2</pre>
<pre>vm.overcommit_ratio=100</pre>
For more information on the implication of this setting, see Linux Overcommit Strategies and Pivotal Greenplum(GPDB)/Pivotal HDB(HDB)

</td></tr>
<tr><td>Operating system/boot disk</td><td>Disk type</td><td>RAID</td><td>Endurance</td><td>Minimum capacity</td><td>Sustained performance IOPS (random 4KB)</td><td>Sustained performance bandwidth (sequential)</td></tr>
</table>

Table 3. Physical server requirements (continued)

<table>
<tr><th>Component</th><th colspan="6">Requirement</th></tr>
<tr><td></td><td>BOSS card</td><td>2*M.2 devices in RAID1</td><td>0.5 DWPD over 5 years or 200PB life span writes</td><td>120GB</td><td>50/20K read/ write</td><td>400/80 [MB/ sec] read/write</td></tr>
<tr><td></td><td colspan="5">NOTE: Due to a limitation, do not install any of the following on the same operating system disk as the MDM: PowerFlex presentation server, PowerFlex Installer, PowerFlex Gateway and VASA processes.</td></tr>
<tr><td>RFCache</td><td colspan="5">• 16 * (cache_size/page_size)
• Commonly-used sizes:</td></tr>
<tr><td></td><td colspan="3">RFcache page size</td><td colspan="2">RFcache memory requirement, if the cache device is 800 GiB</td><td>RFcache memory requirement, if the cache device is 1.6 TiB</td></tr>
<tr><td></td><td colspan="3">64 K</td><td colspan="2">200 MiB</td><td>400 MiB</td></tr>
<tr><td></td><td colspan="3">32 K</td><td colspan="2">400 MiB</td><td>800 MiB</td></tr>
<tr><td></td><td colspan="3">16 K</td><td colspan="2">800 MiB</td><td>1.6 GiB</td></tr>
<tr><td></td><td colspan="3">8 K</td><td colspan="2">1.6 GiB</td><td>3.2 GiB</td></tr>
<tr><td></td><td colspan="3">4 K</td><td colspan="2">3.2 GiB</td><td>6.4 GiB</td></tr>
<tr><td>Disk space</td><td colspan="5">• Disk space formula per node: 1 GB * number of nodes in the system + 1 GB (for log collection)
○ Required path locations: /opt
○ 1 GB for log collection under /tmp (reused every time a log bundle is created)
• For SDC running on Linux or Citrix Hypervisor (XenServer)—1 GiB
NOTE: The minimum operating system size is 10 GiB in a 3 node cluster; 3 GiB is not enough.
• For ESXi—64 GiB is the minimum for a boot disk for VMware topologies; the PowerFlex Storage VM resides on the boot datastore.
Limit: PowerFlex Gateway can't be installed on the same operating system disk as the MDM repository (Dell recommends having them on separate hosts).</td></tr>
<tr><td>Connectivity</td><td colspan="5">One of the following:
• 10/25/40/50/100 GbE network
Dual-port network interface cards (recommended)
Ensure the following:
• There is network connectivity between all components.
• Network bandwidth and latency between all nodes is acceptable, according to application demands.
• Ethernet switch supports the bandwidth between network nodes.
• MTU settings are consistent across all servers and switches.
• The following TCP ports are not used by any other application, and are open in the local firewall of the server:
NOTE: Refer to the Security Configuration Guide for reference to port configuration.
NOTE: You can change the default ports. For more information, see “Changing default ports” in the user documentation.</td></tr>
</table>

NOTE:

If buying a PowerEdge base solution with a PowerFlex software license, be sure to select hardware that is identical to the engineered solution (PowerFlex appliance, PowerFlex rack, VxFlex Ready Node).

PowerEdge configurations that differ from the engineered solution are at risk of encountering issues. Aligning to the engineered solution is critically important when selecting drives on an SDS device.

Agnostic drives should be avoided. Vendor specific drives must be selected in order to ensure the nodes ship with PowerFlex validated drives.

To ensure PowerFlex software compatibility, it is recommended to configure a solution using a PowerFlex base (PowerFlex appliance, PowerFlex rack, VxFlex Ready Node).

Operating system requirements

The following is a list of operating systems supported by this version of PowerFlex.

For the most updated list, see:

- https://elabnavigator.emc.com/vault/pdf/VxFlex_OS.pdf?key=1569877506879 or
- Refer to PowerFlex support matrix at https://www.dell.com/support/home/en-us/product-support/product/scaleio/docs for the supported versions.

  NOTE: Search for v3.5 or relevant version.

Operating system	Requirement
Linux	**NOTE:** Before deploying SDC or RFcache on Ubuntu servers, you must prepare the environment, as described in the *Deploy PowerFlex Guide*. Packages required for all components, all Linux flavors: • numactl • libaio Additional package required for SDC: • which Additional packages required for SDC when installed with SELinux enabled: • policycoreutils (provides utilities for managing SELinux) • selinux-policy (provides SELinux reference policy) • selinux-policy-targeted (provides SELinux taregeted policy) **NOTE:** SELinux will be supported only on operating systems where the status of this feature is production-ready by the operating system vendor\Community. Some operating systems, such as Ubuntu version 20.04, are still in the experimental stage of development. Additional packages required for MDM components: • bash-completion (for SCLI completion) • Latest version of Python 2.X\Python 3.X (for newer OSs such as CentOS\RHEL 8.X) • binutils Additional packages needed by SDS: • smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils Additional packages needed by Diag-tools: • sar • sysstat For supported storage controllers, the following packages are also needed: • DELL: perccli • LSI raid controllers: storcli

Operating system	Requirement
	• HP raid controllers: hpssacli Additional packages required for SDS components that contain NVDIMMs: • ndctl • jq • daxio • libpmem When installing the MDM component on RHEL 6 hosts, set the shared memory parameter in the /etc/sysctl.conf file to at least the following value: kernel.shmmax=209715200. To use this value, type the `sysctl -p` command. To use the secure authentication mode, ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system. • `libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm` • `openssl1-1.0.1g-0.40.1.x86_64.rpm` To use LDAP, ensure that OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: pciutils
Windows	• Only SDC, LIA, and Gateway are supported components on Windows • To install the SDC successfully, ensure that Microsoft Security Update KB3033929 is installed. **NOTE:** It is recommended to apply the latest Microsoft Windows patches. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported) • Visual C++ redistributable 2010 package, 64-bit The Microsoft Windows "Storage Spaces" feature (available from Windows 8.1 and later), is not supported in Microsoft Failover cluster running on PowerFlex volumes.
VMware	HCI\Compute Only (SDC) support: VMware ESXi operating systems: 6.5 U3 or 6.7 U3, managed by vCenter 6.5 or 6.7. **NOTE:** The latest patches of ESXi and vCenter are tested periodically. Refer to https://www.dell.com/support/home/en-us/product-support/product/scaleio/docs and search for OS support matrix file of the relevant release, for example, PowerFlex 3.5.1.x and VxFlex Ready Node 3.5.1.x OS Support Matrix. Compute Only (SDC) support: VMware ESXi operating system: 7.0 (SDC only)

SVM memory requirements

SVM memory requirements.

Component	Memory allocation rules
Memory	PowerFlex RAM (GiB) • SDS FG : 10 GiB + 550 MiB per 1 TiB device size + 100 MiB per device • SDS MG : 5 GiB + 335 MiB per 1TiB device size • MD Read Cache : Total_Drive_Capacity_in_TiB * 8% • MDM : 6.5 GiB • Cloudlink : 4 GiB • Operating System : 1 GiB • LIA : 0.35 GiB

<table>
<tr><th>Component</th><th>Memory allocation rules</th></tr>
<tr><td></td><td>

PowerFlex Installer/PowerFlex Gateway: 8 GiB
VASA: Default 8GiB/Max 16 GiB
SDC: 6 GiB
SDR : 8586 MiB + 550 MiB * (max remote PDs count)

NOTE: SDR SVM support is only available as part of a PowerFlex appliance or the PowerFlex IR system managed by PowerFlex Manager .

SDR memory requirement for HC systems (SVMs): 12GB

NOTE: For certain system configurations, the SDS needs to be configured to be able to use more than one Non-Uniform Memory Access (NUMA) domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and therefore, by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0. If NUMA 0 (usually half the total memory) is less than the memory required by the SDS, you need to allow the SDS access to the memory in the other NUMA. For more information, see the explanation and configuration procedures for "Configure the SDS for optimal memory usage" in "Post-deployment tasks" in the Deploy Dell PowerFlex .

Non-Volatile RAM (NVDIMM):

Node Storage Capacity < 49 TiB : NVDIMM size 32 GiB
Node Storage Capacity between [49 TiB, 96 TiB] : NVDIMM size 64 GiB
Node Storage Capacity > 96 TiB : NVDIMM size 96 GiB

NOTE: PowerFlex does not support using swap on Storage only and SVM nodes. From version 3.0 and above, this should be the default setting.

Using swap with PowerFlex has the following impact:

Slows the performance of the system
Impacts I/O on the system disks which is used for swap. This may cause other functions that use the system disk to have slow I/Os such as:

MDM repository related operations
PowerFlex logs written to disk.

Disable swap and set <code>vm.overcommit</code> to the following on nodes that have MDM and SDS process on them in the <code>in/etc/sysctl.conf</code> file:
<pre>vm.overcommit_memory=2</pre>
<pre>vm.overcommit_ratio=100</pre>

</td></tr>
<tr><td>RFcache</td><td>

16 * (cache_size/page_size)
Commonly-used sizes:

<table>
<tr><th>RFcache page size</th><th>RFcache memory requirement, if the cache device is 800 GiB</th><th>RFcache memory requirement, if the cache device is 1.6 TiB</th></tr>
<tr><td>64 K</td><td>200 MiB</td><td>400 MiB</td></tr>
<tr><td>32 K</td><td>400 MiB</td><td>800 MiB</td></tr>
<tr><td>16 K</td><td>800 MiB</td><td>1.6 GiB</td></tr>
<tr><td>8 K</td><td>1.6 GiB</td><td>3.2 GiB</td></tr>
<tr><td>4 K</td><td>3.2 GiB</td><td>6.4 GiB</td></tr>
</table>
</td></tr>
<tr><td>Base SVM</td><td>

350 MiB

</td></tr>
<tr><td>Tiebreaker MDM</td><td>

50 MiB

</td></tr>
</table>

PowerFlex presentation server requirements

See PowerFlex Support Matrix for PowerFlex GUI management server requirements:

Component	Requirement
CPUs	At least two CPU cores
Memory	• Small or normal scale clusters: 4 GB • Large-scale clusters: 6 GB
System disk size	1.25 GB for the process+351 MB for logs Total=1.6 GB
Supported operating systems for PowerFlex	See https://www.dell.com/support/home/en-us/product-support/product/scaleio/docs for updated list (PowerFlex Support Matrix) of supported operating systems for the current version of PowerFlex.
Supported browsers	• Firefox® • Google Chrome™ • Microsoft Edge® • Internet Explorer® • Safari® • Opera
Oracle JRE	• 1.8 build 301 • Download page for Linux file: jre-8u301-linux-x64.rpm • Download page for Windows x86_64: jre-8u301-windows-x64.exe
OpenJDK	• JDK 8 build 302.b08 ○ Main download page: https://pkgs.org/download/java-1.8.0-openjdk-headless ○ el7: https://buildlogs.centos.org/centos/7/isos/x86_64/ ○ el8: https://buildlogs.centos.org/8-stream/nfv/x86_64/ • JDK build 11.0.12.0.7 ○ Main download page: https://pkgs.org/download/java-11-openjdk-headless(x86-64) ○ el7: https://buildlogs.centos.org/centos/7/isos/x86_64/ ○ el8: https://buildlogs.centos.org/8-stream/nfv/x86_64/ **NOTE:** Windows is not supported for OpenJDK.
Connectivity	Other applications must not use the following TCP ports, and the TCP must be open in the local firewall of the server: 80 and 443 (or 8080 and 8443). **NOTE:** The PowerFlex GUI management server and the PowerFlex Gateway both use port 8443. In cases where the PowerFlex GUI management server and PowerFlex Gateway are deployed on the same server (not recommended), define port 9443 for the PowerFlex GUI management server. For more information about changing the port and configuring the firewall, see the *Deploy Dell PowerFlex* .

PowerFlex Gateway server requirements

Refer to PowerFlex Support Matrix for PowerFlex Gateway server requirements:

Component	Requirement
Supported operating systems	• Refer to https://elabnavigator.emc.com/eln/modernHomeSSM for list of supported operating systems. • "rpmlib" version should be 4.6.0-1 or higher Every server requires 2 cores and a minimum of 3 GiB available RAM.

Component	Requirement
CPUs for PowerFlex Installer/ PowerFlex Gateway	2
Memory	8 GiB
Operating system/boot disk	Required size: 10GB (minimum operating system disk size) + number of nodes*1GB This is required by the PowerFlex Gateway to allow collecting logs from all nodes. **NOTE:** Due to a limitation, do not install any of the following on the same operating system disk as the MDM: PowerFlex presentation server, PowerFlex Installer, PowerFlex Gateway and VASA processes.
Connectivity	The following TCP ports are not used by any other application, and are open in the local firewall of the server: 80 and 443 (or 8080 and 8443). **NOTE:** The PowerFlex GUI management server and the PowerFlex Gateway both use port 8443. In cases where these will be deployed on the same server, define a different port for the PowerFlex GUI management server. For more information about changing default ports, see "Changing default ports" in the user documentation.
Supported web browsers	• Firefox, version 65.0.1, or later • Chrome, version 72, or later
Oracle JRE	• 1.8 build 301 • Download page for Linux file: jre-8u301-linux-x64.rpm • Download page for Windows x86_64: jre-8u301-windows-x64.exe
OpenJDK	• JDK 8 build 302.b08 ○ Main download page: https://pkgs.org/download/java-1.8.0-openjdk-headless ○ el7: https://buildlogs.centos.org/centos/7/isos/x86_64/ ○ el8: https://buildlogs.centos.org/8-stream/nfv/x86_64/ • JDK build 11.0.12.0.7 ○ Main download page: https://pkgs.org/download/java-11-openjdk-headless(x86-64) ○ el7: https://buildlogs.centos.org/centos/7/isos/x86_64/ ○ el8: https://buildlogs.centos.org/8-stream/nfv/x86_64/ **NOTE:** There is no Windows support for OpenJDK.
Other	• For a PowerFlex Gateway server on a Windows node, the Windows Management Instrumentation service must be enabled on the PowerFlex Installer server and on all Windows PowerFlex nodes. • Do not install the PowerFlex Gateway on a server on which RFcache will be enabled or on which SDC will be installed. • The PowerFlex Gateway server must have connectivity to all the nodes that are being installed. If you are using separate networks for management and data, the server must be able to communicate with both networks.

Other requirements

This section lists additional requirements for PowerFlex.

PowerFlex requires that you use a minimum of three SDS servers, with a combined free capacity of at least 720 GB. These minimum values are true per system and per Storage Pool.

Device encryption software

Software	Support
CloudLink	6.9 and 7.0
PowerFlex SRA (Storage Replication Adapter)	Supported on SRM (Site Recovery Manager) 8.2 and 8.3

VASA provider server requirements

Component	Requirement
CPUs	8 cores
Memory	Small/normal scale clusters: 8 GiB Large scale clusters: 16 GiB
Supported operating systems for PowerFlex	Refer to PowerFlex Support Matrix for full list of supported OS's
Oracle JRE	• 1.8 build 301 • Linux download file: jre-8u251-linux-x64.rpm • Download links for Windows x86_64: https://github.com/ojdkbuild/ojdkbuild#downloads-for-windows-x86_64
OpenJDK	• JDK 8 build 302.b08 • Linux file name: OpenJDK8U-jre_x64_linux_8u252b09.tar.gz • For guidelines on how to get and install OpenJDK for Linux Distro, refer to: https://openjdk.java.net/install/. • Download links for Windows x86_64: https://github.com/ojdkbuild/ojdkbuild#downloads-for-windows-x86_64 • Windows download file: java-1.8.0-openjdk-1.8.0.252-2.b09.ojdkbuild.windows.x86_64.msi

Before you begin

Ensure that:

- Each host that runs the VASA provider has sufficient memory. The minimum required amount is 8 GB, though 16 GB is recommended.
- When using HCI configuration with VASA provider, the SVM needs 8GB of memory.
- Oracle JRE's latest build must be installed on every host that will run the VASA provider. The correct Java executables must be found using the PATH environment variable.
- The DNS server must be configured, and for every VASA host, a DNS record should be created. Ensure that all VASA hosts have a valid DNS resolution from all ESXi hosts and the vCenter Server, and that the vCenter Server has a valid DNS resolution from all VASA hosts.
- Each VASA VM's hostname must match the VASA FQDN (the output of commands `hostname` and `hostname -f` should be the same).
- The VASA VM's hostname consists only of ASCII letters, digits, and hyphens, and must start with a letter character.
- vCenter and ESXis must see the VASA VM host by the same domain name as `hostname --fqdn` performed on the host
- All components of your environment, including ESXis and vCenters, have their time synchronized.
- SDC should be installed on each ESXi which will consume PowerFlex vVols.
- PowerFlex is deployed.

II

Architecture

This section describes the architecture of the PowerFlex system.

3

Architecture

The following topics describe the different components of the PowerFlex system.

System

PowerFlex is based on a hardware and a software component.

Hardware

This section outlines the PowerFlex hardware components.

In general, hardware can be the existing application servers used by the datacenter, or a new set of nodes (if, for example, you want to dedicate all nodes solely for the purpose of running the PowerFlex SAN storage system).

The PowerFlex rack system includes a full datacenter solution including: rack, switches, server nodes , power and cabling. Built for capacity, performance, or compute per the specific configuration.

- *Nodes*

  Nodes, or servers, are the basic computer unit used to install and run PowerFlex. They can be the same servers used for the applications (server convergence), or a dedicated cluster. In any case, PowerFlex is hardware-agnostic, and therefore, aside from performance considerations, the type of server is inconsequential.

- *Storage Media*

  The storage media can be any storage media, in terms of the type (HDD, SSD, NVMe SSD or PCIe flash cards) and anywhere (external). PowerFlex appliance and VxFlex Ready Node are additional offerings that have PowerFlex engineered solution.

Software

The PowerFlex virtual SAN software consists of a Meta Data Manager (MDM), a Storage Data Server (SDS) and a Storage Data Client (SDC).

The PowerFlex virtual SAN consists of the following software components:

- Meta Data Manager (MDM)

  Configures and monitors PowerFlex. The MDM can be configured in redundant cluster mode, with three members on three servers or five members on five servers, or in single mode on a single server.

  NOTE: It is not recommended to use single mode in production systems, except in temporary situations. The MDMs contains all the metadata required for system operation. single mode has no protection, and exposes the system to a single point of failure.

- Storage Data Server (SDS)

  Manages the capacity of a single server and acts as a back-end for data access. The SDS is installed on all servers contributing storage devices to PowerFlex. These devices are accessed through the SDS.

- Storage Data Client (SDC)

  A lightweight device driver that exposes PowerFlex volumes as block devices to the application that resides on the same server on which the SDC is installed.

- Storage Data Replication (SDR)

A component that acts as a proxy between the SDS and SDC that must be installed for replication. Journal capacity percentage must be allocated in order to add SDRs to the system. A minimum of two SDRs is required on each replication system. Best practice is to have SDR and SDS on one server. You can add, remove and rename SDRs. Depending on the role specified for the SDR, that is what is used to communicate with the SDSs on the peer system.

NOTE: Refer to SDR limits for max supported SDR count.

Depending on the desired configuration (described later), the software components are installed on the server and give rise to a virtual SAN layer exposed to the applications that reside on the servers.

The MDM cluster

The MDM, functions as the monitoring and configuration agent of PowerFlex and is used mainly for management. A multi-MDM environment consists of one primary MDM, while the others function as secondary or tiebreaker.

The MDM is mainly used for management which consists of migration, rebuilds, and all system-related functions. No I/Os run through the MDM.

To support high availability, three or five instances of MDM run on different servers. In a multi-MDM environment, one MDM is given the primary role, and the others act as secondary or tiebreaker MDMs.

The MDM cluster comprises a combination of primary MDM, secondary MDMs, and tiebreaker MDMs. There is also the standby MDM which is not a part of the cluster.

The following terms are relevant to the MDM, the building blocks of the MDM cluster:

- MDM

  Any server with the MDM package installed. An MDM can be given a manager or a tiebreaker (default) role, during installation. This role can't be changed later without reinstalling the MDM. MDMs have a unique MDM ID, and can be given unique names.

  Before the MDM can be part of the cluster, it must be promoted to a standby MDM.

- Standby MDM and tiebreaker

  An MDM and a tiebreaker can be added to a system as a standby. Once added, the standby MDM or tiebreaker is attached, or locked, to that specific system.

  A standby MDM can be called on to assume the position of a manager MDM or tiebreaker MDM according to how it is installed, when it is promoted to be a cluster member.

- Manager MDM

  An MDM that can act as a primary or a secondary in the cluster. Manager MDMs have a unique system ID, and can be given unique names. A manager can be a standby or a member of the cluster.

  In PowerFlex documentation, “MDM” refers to a manager, unless specified otherwise.

- Tiebreaker MDM

  An MDM whose sole role is to help determine which MDM is the primary. A tiebreaker can be a standby or a member of the cluster. A tiebreaker MDM is not a manager.

  In a 3-node cluster, there is one tiebreaker; in a 5-node cluster, there are two tiebreakers. This ensures that there are always an odd number of MDMs in a cluster, which guarantees that there is always a majority in electing the primary.

The following terms are relevant to the MDM cluster, specifically:

- Primary MDM (previously called master MDM)

  The MDM in the cluster that controls the SDSs and SDCs. The primary MDM contains and updates the MDM repository, the database that stores the SDS configuration, and how data is distributed between the SDSs in the system. This repository is constantly replicated to the secondary MDMs, so they can take over with no delay.

  Every MDM cluster has one primary MDM.

- Secondary MDM (previously called slave MDM)

  An MDM in the cluster that is ready to take over the primary MDM role if ever necessary.

In a 3-node cluster, there is one secondary MDM, thus allowing for a single point of failure. In a 5-node cluster, there are two secondary MDMs, thus allowing for two points of failure. This increased resiliency is a major benefit to enabling the 5-node cluster.

- Replica

  An MDM that contains a replica of the MDM repository. This includes the primary MDM and any secondary MDMs in the MDM cluster.

The following table describes the available cluster modes:

Table 4. MDM cluster modes

Cluster mode	Members	Description
3-node (default)	• Primary MDM • Secondary MDM • Tiebreaker	3-node cluster has two copies of the repository, thus can withstand one MDM cluster failure.
5-node	• Primary MDM • Two secondary MDM • Two tiebreaker	5-node cluster has three copies of the repository, thus can withstand two MDM cluster failure.
Single-node	• Primary MDM	Single-node cluster has only one copy of the repository, thus it cannot withstand failure. It is not recommended to use Single Mode in production systems.

In addition to the cluster members, you can prepare standby managers and tiebreaker nodes, for a total of thirteen cluster and standby MDMs.

The MDM cluster IP address limit is 16 IP addresses, which includes all cluster members (primary, secondary, standby primary, and standby secondary).

The following figure illustrates a 5-node MDM cluster:

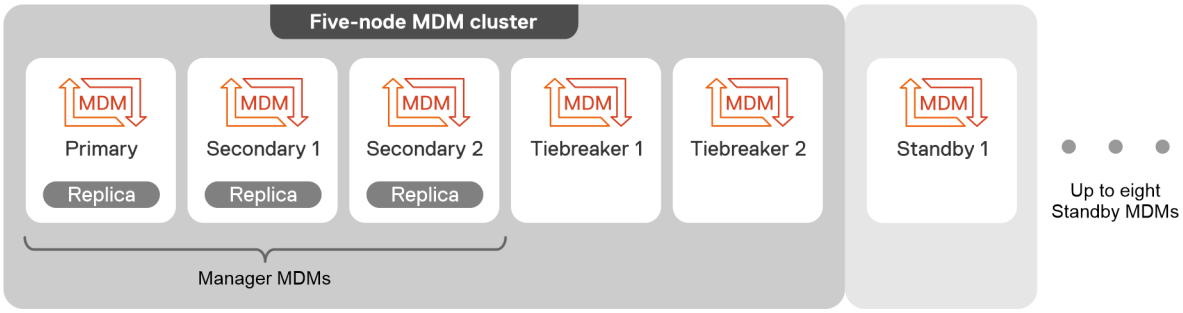


Figure 1. 5-node MDM cluster

All members of the MDM cluster have the same MDM package installed on them.

Before a server makes its way into the MDM cluster, it must follow the following path:

1. Install the MDM package on the server.

   During the installation, you determine if the server will be a manager or a tiebreaker (default).

2. Promote the server to standby status, either as a manager or as a tiebreaker according to how it was installed.
3. Add the standby server to the MDM cluster. A manager, once entered into the cluster can take on the primary or secondary state.

MDM cluster creation is done automatically when deploying a system with any of the automated deployment tools.

The following list includes best practices for adding MDMs to a multiple-rack system:

- For a large system with multiple racks, the MDM cluster members should be distributed in separate racks.
- The MDM network requirements described in the "Networking" section, must also apply to the network connections between the racks.
- You can ensure higher resiliency when the rack is shut down for planned\unplanned reasons, especially if Fault Sets are used. Refer to section on "Fault Sets".
- For simultaneous upgrade of multiple nodes in different racks, it is important to take note of the nodes which have MDMs. Do not simultaneously reboot\shut down the nodes which have MDMs. To ensure users do not shut down the MDMs by mistake, mark them physically with stickers saying "Do not shut down: MDM installed here", or add a meaningful prefix\suffix to the hostname of the server where MDM is installed. For example: *Rack22U32-MDM* You can also make sure a different node in the rack is used per server as shown in the table:

Node level	Rack number	MDM role
Top node	1	MDM
Bottom node	2	MDM
Middle node	3	TB

Virtual IP Address

Virtual IP addresses can be defined for the MDM cluster.

Up to four virtual IP addresses can be defined for the MDM cluster. SDCs are then mapped to the MDM cluster's virtual IP addresses, instead of to static MDM IP addresses. MDMs are sometimes switched during normal operation of the cluster, and the virtual IP address will always be mapped to the active MDM. The use of virtual IP addresses simplifies maintenance procedures on the MDM cluster, because system components communicate via the virtual IP addresses. Therefore, SDCs do not need to be reconfigured when a server hosting an MDM is replaced.

In new installations in Linux environments, the MDM cluster's virtual IP address can be added and mapped using the PowerFlex Installer CSV file. In VMware environments, virtual IP addresses are mandatory, and configuration is performed using the PowerFlex VMware Installation Wizard, in the Configure SVM stage. The REST API can also be used to add virtual IP addresses to the cluster. In all cases, a virtual IP NIC placeholder must be mapped to each virtual IP address. Ensure that there are NICs available for this purpose.

Existing systems may be extended to include additional MDMs to a cluster. The new MDMs should be mapped to the existing virtual IP addresses.

If virtual IP addresses need to be modified, you must use the CLI or the REST API (not the PowerFlex Installer or the vSphere plug-in), and it must be done with extreme caution.

All SDCs will require reconfiguration, to reflect the changes made to the MDM cluster. Otherwise, the SDCs will not be able to communicate with the MDM cluster, and volumes will not be accessible.

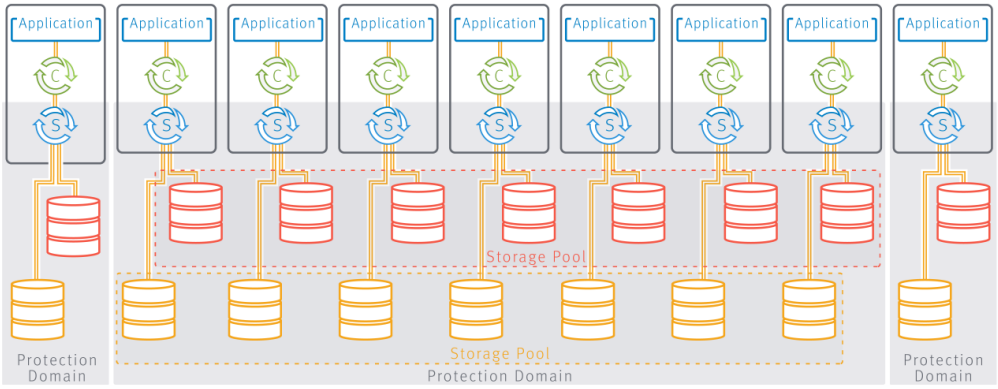
Storage definitions

When configuring PowerFlex, you should take the following concepts into account: Protection Domains, Storage Pools, and Fault Sets. Together, these elements link the physical layer to the virtual storage layer.

Protection Domains

A Protection Domain is a logical entity that consists of a group of SDSs that backs up each other. Protection Domains can be added during installation and modified post-installation.

A Protection Domain is a logical entity that contains a group of SDSs that provide backup for each other. Each SDS belongs to one (and only one) Protection Domain. Thus, by definition, each Protection Domain is a unique set of SDSs. In the following image there are three Protection Domains. The one in the middle (fully depicted) consists of seven SDSs, each with two storage devices.



The maximum recommended number of nodes in a Protection Domain is 128. This enables the following:

- optimal performance
- reduction of theoretical *mean time between failure* issues
- ability to sustain multiple failures in different Protection Domains

You can add Protection Domains during installation. In addition, you can modify Protection Domains post-installation with all the management clients (except for OpenStack) .

Practical Aspects to Implementing Nodes in the Protection Domain

Mix Different Node Types in the Protection Domain

When mixing different node types in the Protection Domain the following impacts occur:

- The slowest node will determine the speed of all other nodes in the Protection Domain.
- The storage pool depends on the association of disks in a node to a single storage pool or multiple storage pools.

Expand existing DellPowerEdge R630/R730 and Dell PowerEdge R640/R740xd/R840 PowerFlex systems with Dell PowerEdge R650/R750/R6525 PowerFlex nodes

Although PowerFlex systems do not enforce any restrictions, the best practice is to have the same node configuration throughout the entire Protection Domain. Overall system performance and or availability might be severely impacted when nodes with drastically different I/O throughput, CPU, and capacity are mixed together.

Expand existing Dell PowerEdge R630/R730 or Dell PowerEdge R640/R740xd/R840 PowerFlex protection domains with Dell PowerEdge R650/R750/R6525 PowerFlex nodes

PowerFlex manager does not currently enforce any restriction against mixing nodes in Protection Domains. The best practice is to expand a Protection Domain with the same hardware (CPU, memory speed, memory capacity, drive type, drive capacity, and networking ports). For dissimilar hardware, such as Dell PowerEdge R640/R740xd/R840 and Dell PowerEdge R650/R750/R6525, the best practice is to expand the cluster with new Protection Domains.

NOTE: When mixing different generations of PowerFlex nodes the performance is dictated by the lowest performing component, for example Dell PowerEdge R650/R750/R6525 nodes performs at the speed of Dell PowerEdge R640/R740xd/R840 if mixed with Dell PowerEdge R640/R740xd/R840 nodes in the same Protection Domain.

An evergreen use case exists when adding Dell PowerEdge R650/R750/R6525 hardware into existing Dell PowerEdge R640/R740xd/R840 systems. The workloads move to new nodes then the old nodes are removed. From a PowerFlex Manager perspective you can mix Dell PowerEdge R640/R740xd/R840 and Dell PowerEdge R650/R750/R6525 are using 25 G networking.

PowerFlex Manager does not configure VMware Enhanced vMotion Capability (EVC).

Fault Sets

A Fault Set defines a group of SDSs that are more inclined to go offline simultaneously. Defining a Fault Set configures PowerFlex to mirror the devices in the set.

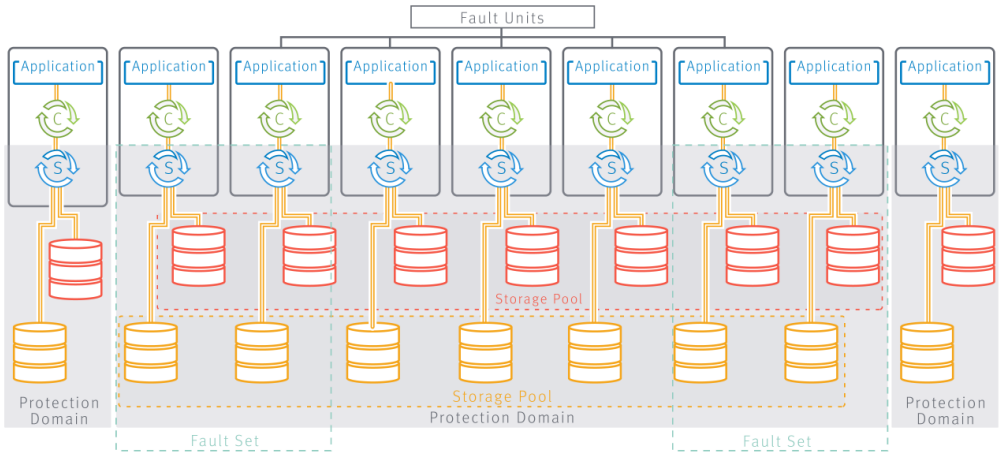
A Fault Set is a logical entity that contains a group of SDSs within a Protection Domain, that have a higher chance of going offline together, such as a group of SDSs that are all powered in the same rack. By grouping them into a Fault Set, you are configuring PowerFlex to mirror the data for all devices in this Fault Set. The mirroring should take place on SDSs that are outside of this Fault Set.

When defining Fault Sets, we refer to the term fault units, where a fault unit can be either a Fault Set, or an SDS not associated with a Fault Set (you may think of it as a Fault Set of a single SDS).

There must be enough capacity within at least three fault units to enable mirroring.

If Fault Sets are defined, you can use any combination of fault units, for example:

- SDS1, SDS2, SDS3
- FS1, SDS1, SDS2
- FS1, FS2, SDS1
- FS1, FS2, FS3



To use Fault Sets, you must work in the following order:

1. Ensure that a Protection Domain exists, or add a new one.
2. Ensure that a Storage Pool and Fault Sets (minimum of three fault units) exist, or add new ones.
3. Add the SDS, designating the Protection Domain and Fault Set, and at the same time, adding the SDS devices into a Storage Pool.

   The automated deployment and installation tools follow this order automatically.

   You can only create and configure Fault Sets before adding SDSs to the system, and configuring them incorrectly may prevent the creation of volumes. An SDS can only be added to a Fault Set during the creation of the SDS.

   You define Fault Sets and add SDSs to them during installation, using the following management tools:

- PowerFlex Installer
- CLI
- REST
- vSphere plug-in

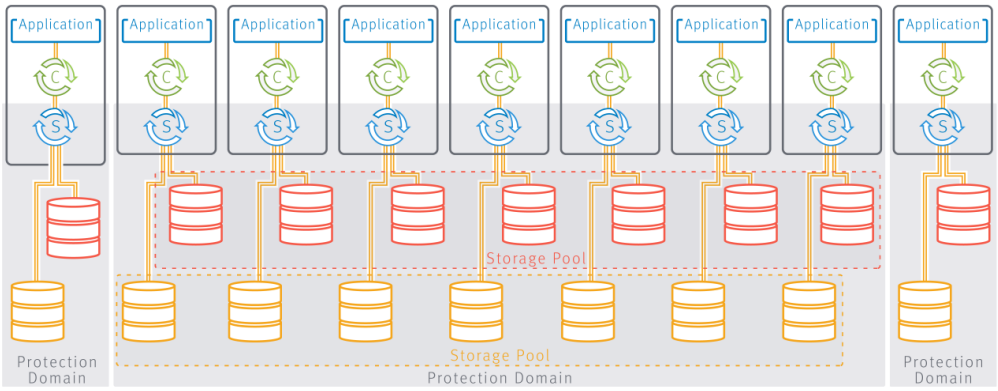
You can also add Fault Sets when adding SDS nodes after initial installation.

Storage Pools

A Storage Pool is a set of physical storage devices in a Protection Domain. A volume is distributed over all devices residing in the same Storage Pool. You can define a Magnetic Storage Pool (for HDDs) or a High Performance Storage Pool (for SSDs). Storage Pools support medium or fine granularity data layouts and allow enabling or disabling zero padding.

Storage Pools allow the managing of different storage tiers in PowerFlex. A Storage Pool is a set of physical storage devices in a Protection Domain. Each storage device belongs to one (and only one) Storage Pool. The figure "Protection Domains and Storage Pools" depicts two Storage Pools.

When a volume is configured over the virtualization layer, (see "SAN virtualization layer"), it is distributed over all devices residing in the same Storage Pool. Each volume block has two copies located on two different Fault Units. This allows the system to maintain data availability following a single-point failure. Two network failures render all the domain in a state where IO errors are possible on any storage pool



You must assign a media type setting to each Storage Pool. Supported types are: HDD, SSD and Transitional (allows for migration flows).

If all SDSs in a Protection Domain have two physical drives associated with them—one HDD, and the other SSD— you should define two Storage Pools:

- Magnetic Storage Pool

  Consists of all HDDs in the Protection Domain

- High performance Storage Pool

  Consists of all SSDs used for storage purposes in the Protection Domain

  NOTE: Mixing different types of SSDs is not recommended and creating a separate Storage Pool for each type is the recommended best practice. For example: SAS SSD, SATA SSD, NVMe SAS SSD.

  NOTE: PowerFlex might not perform optimally if there are large differences between the sizes of the fault units in the same Storage Pool. For example, if one device has a much larger capacity than the rest of the devices, performance may be affected. After adding devices, you can define how much of the device capacity is available to PowerFlex by using the SCLI `modify_sds_device_capacity` command.

Storage Pools support the following data layouts for HDD or SSD media:

- Medium Granularity (MG): space allocation occurs at 1MB units
  - suppports HDD and SSD media
  - includes persistent checksum for data integrity
- Fine Granularity (FG):
  - requires SSD media and NVDIMM for acceleration
  - space allocation occurs at 4KB units
  - includes persistent checksum for data integrity
  - supports data compression which reduces the size of data stored on the disk
  - supports thin-provisioned, zero-padded volumes

NOTE: FG and MG storage pools can both exist in a single SDS. You can also migrate volumes across the two layouts.

Each Storage Pool can work in one of the following modes:

- Zero-padding enabled

  Ensures that every read from an area previously not written to returns zeros. Some applications might depend on this behavior. Furthermore, zero padding ensures that reading from a volume will not return information that was previously deleted from the volume.

  This behavior incurs some performance overhead on the first write to every area of the volume since the area needs to be filled with zeros first.

FG is always zero padded.

- Zero-padding disabled (default only for MG)

  A read from an area previously not written to will return unknown content. This content might change on subsequent reads.

Zero padding must be enabled if you plan to use any other application that assumes that when reading from areas not written to before, the storage will return zeros or consistent data.

You can add Storage Pools during installation. In addition, you can modify Storage Pools post-installation with most of the management clients.

NOTE: The zero padding policy cannot be changed after the addition of the first device to a specific Storage Pool.

Storage Pool spare capacity

Storage Pool spare capacity is the space used by the PowerFlex system in case of failure or maintenance operations of a node. The minimum allocation size is equal to the space of one node in the Storage Pool (assuming all nodes are the same size)

Example:

There are ten nodes, each with 10 TB part of the Storage Pool. Since 50% is allocated to Raid 1, the actual capacity in the Storage Pool is 50 TB, leaving 5 TB worth of usable space for each node.

The recommended spare capacity is 10% = 5 TB.

If a node is placed in maintenance mode, the spare space is used to replace the node that is currently down.

This is mandatory for the upgrade operation that involves placing SDSs in maintenance mode during upgrade and similar maintenance use cases.

A higher setting of spare capacity is supported. It enables more than a single node to be down at the same time with one limitation, each node fails at a separate time and rebuild finished (or node finished entering persistent maintenance mode). It doesn't support double node failure at the same time.

For example: A node was placed in PMM to replace a hardware component and during this time of maintenance, a second failure occurred.

The table below shows the spare capacity % recommended for one or two nodes that may be down.

Example of spare capacity sizing:

Number of nodes in Storage Pool	Spare capacity % allowing one node to be down	Spare capacity % allowing two nodes to be down
3	34%	Not supported
10	10%	20%
20	5%	10%
30	3%	6%

Acceleration Pools

Acceleration Pools enable RFCache of SSD or NVMe devices to accelerate an HDD storage pool. It also enables NVRAM required for Fine Granularity (FG) storage pools.

RFCache Acceleration Pools features and requirements:

- RFCache feature consumes the entire device, which prevents other features being used
- You can only have a single RFCache Acceleration Pool per Protection Domain
- NDU creates a default RFCache Acceleration Pool
- You can disable/enable RFCache via PD, SP, and SDS
- A storage device benefits from RFCache acceleration if RFCache is enabled on all three levels and if the SDS actually contains RFCache devices
- NVMe disks can't be used with RFCache as an acceleration device

NVRAM Acceleration Pools features and requirements:

- Must reside on NVDIMM devices

- The NVDIMM must be mapped to a DAX Device
- NVRAM Acceleration Pools cannot be disabled
- Migration to a different NVRAM pool is not supported

Limitation: NVDIMM failure on SDS node causes failure of all node's SSDs in the FG storage pool.

Data layout

PowerFlex offers a space efficient store layout with Fine Granularity (FG)

With FG, since space allocation occurs at 4KB, the data stored on the disk is significantly reduced. Space allocation for Medium Granularity (MG) occurs at 1MB.

For FG, you must have NVDIMM configured as a DAX device. After NVDIMM is all set up in your environment, you need to configure an NVDIMM Acceleration Pool (for more information on Acceleration Pool, see "Acceleration Pools") and then you can select it in the Storage Pool (for more information on Storage Pool, see "Storage Pools").

FG enables data compression which allows for faster reads and writes. Data-compression is not supported for Medium Granularity (MG). It supports thin provisioned, zero padded volumes. Zero padded is enabled by default for FG and can't be disabled. Zero padding is disabled for MG.

FG supports persistent checksum for data integrity. For more information on persistent checksum, see "Checksum protection".

Volumes

You must add and configure volumes before enabling applications to access them.

The adding and mapping volume process is necessary, as part of the getting started process, before applications can access the volumes. In addition, you may create additional volumes and map them as part of the maintenance of the virtualization layer.

You can configure the caching option when creating the volumes, or you can change the Read RAM Cache feature later. If you want to enable the caching feature, ensure that the feature is also enabled in the backend of the system, for the corresponding Storage Pool and SDSs. For more information, see "Change Read RAM Cache volume settings".

Define volume names according to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID via a query, or via the object's property sheet in the PowerFlex GUI. It is highly recommended to give each volume a meaningful name associated with its operational role.

Naming

Giving meaningful names to PowerFlex objects can simplify volume-related actions.

It is recommended to name all PowerFlex objects with meaningful names. This will make it easier when defining volumes, associating them with applications, etc.

From the previous example, the Storage Pools can be named "Capacity_Storage" and "Performance_Storage," which allows you to identify the different tiers.

As for Protection Domains, one example would be separating the SDSs used by the finance department from those used by the engineering department. This segregation of different departments is very beneficial in many aspects (security being one of them). Thus, one might name the domains "Financial-PD" and "Engineering-PD."

The Fault Sets could be called "FS_Rack01" and "FS_Rack02."

Replication

PowerFlex enables native asynchronous replication for storage-only and HCI configurations.

Replication can be used to quickly and easily recover from a physical or logical disaster, to migrate data, to test data at a remote site, or to offload backup. PowerFlex's implementation is designed to allow a sub-minute RPO reducing the data-loss to minimal in case of disaster recovery. As with all other PowerFlex data services, the replication is elastic, can scale online by adding or removing nodes, flexible, , and easy to manage. It enables instant test and failover operations.

The following terms and concepts define the replication architecture and its components:

Replication peer systems

Replication occurs between two PowerFlex systems, connected by WAN, and designated as peer systems.

Replication is always defined in the scope of a Protection Domain. All the objects that participate in the replication are contained in the Protection Domain including the volumes in a replication consistency group (RCG). The journal capacity from the Storage Pool in a Protection Domain is shared among all the RCGs in the Protection Domain. The SDRs in the Protection Domain manage I/Os directed to replicated volumes within the Protection Domain.

SDR

Storage Data Replicators (SDR) are responsible for processing all I/Os of replication volumes.

The source SDRs processes all application I/Os of replicated volumes. At the source, application I/Os send the SDC to the SDR. The I/Os get sent to the target SDRs and stored in their journals. The target SDRs' journals apply the I/Os to the target volumes. A minimum of two SDRs get deployed at both the source and target systems to maintain high availability. If one SDR fails, the MDM directs the SDC to send the I/Os to an available SDR.

There is a limit to the ratio of SDRs in a source protection domain to the SDRs in its target protection domain. The maximum ratio of peer SDRs is 1:3, and The maximum ratio of SDRs in maintenance is 1:6. `add_sdr` and `remove_sdr` commands fail if the operation results in an unsupported ratio.

`add_replication_consistency_group` command fails, If the SDRs ratio exceeds the allowed value as a result of a failure or planned maintenance. There is no impact on existing replication consistency groups and replication operations.

Journal

The journal is a component of the SDR.

The journal stores the data at the source before it is sent to the destination. At the destination, the journal stores the data before it is applied to the destination volumes. At the source, application I/Os are sent by the SDC to the SDR. The SDR packages I/Os in bundles and sends them to the target journal. Once the I/Os are stored at the destination journal, they are cleared from the source journal. Once the I/Os are applied to the target volumes, they are cleared from the destination journal.

SDR journal capacity

You should consider several factors when allocating journal capacity.

Journal capacity is defined as a percentage of the total storage capacity in the Storage Pool and must equal at least 28 GB per SDR. In general, journal capacity should be at least 5% of replicated usable capacity in the Protection Domain, including volumes used as source and targets. It is important to assign enough storage capacity for the replication journal.

The amount of capacity needed for the journal is based on the following factors:

- Minimal requirements — 108 GB multiplied by the number of SDR sessions. The number of SDR sessions is equal to the number of SDRs plus one. The extra SDR session is to ensure that a new session can be allocated for an SDR during an NDU.
- The capacity needed to sustain an outage — application WAN bandwidth multiplied by the planned WAN outage. In general, journal capacity in the Protection Domain should be at least 5% of the application pool. If the application has a heavy I/O load, larger capacity should be used. Similarly, if a long outage is expected, a larger capacity should be allocated. If there are replicated volumes in more than one Storage Pool in the Protection Domain, this calculation should be repeated for each Storage Pool, and the allocated journal capacity in the Protection Domain must at least equal the sum of the size per application pool.

Use the following steps to calculate exactly how much journal capacity to allocate:

1. Select the Storage Pool(s) from which to allocate the journal capacity. The journal is shared between all of the replicated RCGs in the Protection Domain. Journal capacity should be allocated from Storage Pools as fast as (or faster than) the Storage Pool of the fastest replicated application in the Protection Domain. It should use the same drive technology and about the same drive count and distribution in nodes.
2. Consider the minimal requirements needed (28 GB multiplied by the number of SDR sessions) and the capacity needed to sustain an outage. Journal capacity will be at least the maximum of these two factors.
3. Take into account the expected outage time. The minimal outage allowance is one hour, but at least three hours are recommended.
4. Calculate the journal capacity needed per application: maximal application throughput x maximum outage interval.
5. Since journal capacity is defined as a percentage of Storage Pool capacity, calculate the percentage of capacity based on the previously calculated needs.

For example:

- An application generates 1 GB/s of writes.
- The maximal supported outage is 3 hours (3 hours x 3600 seconds = 10800 seconds).
- The journal capacity needed for this application is 1 GB/s x 10800 s = ~10.547 TB.
- Since the journal capacity is expressed as a percentage of the Storage Pool capacity, divide the 10.547TB by the size of the Storage Pool, which is 200TB: 100 x 10.547TB/200TB = 5.27% Round this up to 6%.
- Repeat this for each application being replicated.

When a Protection Domain has several Storage Pools and several replicated applications, the journal capacity should be calculated as in the example above and the capacity can be divided among all the Storage Pools (provided they're fast enough). For higher availability, the journal capacity should be allocated from multiple Storage Pools.

NOTE: When Storage Pool capacity is critical, capacity cannot be allocated for new volumes or for expanding existing volumes. This behavior must be taken into account when planning the capacity available for journal usage. The volume usage must leave enough capacity available in the Storage Pool to allow provisioning of journal volumes. The plan should account for the Storage Pool staying below critical capacity even when the journal capacity is almost fully utilized.

It is important to note that since journal capacity is defined as a percentage of the total storage capacity in the Storage Pool, increasing the total storage capacity by adding devices will increase the journal capacity. Similarly, if you decrease the total storage capacity by removing devices from the Storage Pool, the journal capacity will automatically decrease.

Replication consistency group

The replication consistency group (RCG) is a logical container for volumes whose application data need to be replicated consistency to each other.

For example, if you replicate a database and its transaction log, you need the remote copy of each of the database volumes to reflect an image from the same point in time. You can achieve this by placing all of the database volumes in one RCG.

The following replication attributes are defined in the RCG:

- RPO (Recovery Point Objective)—The maximum data loss (in units of time) that you are willing to lose. When you set the RPO, this is the goal of replication.
- Consistency mode—The RCG consistency mode defines the way the data is applied at the destination. When the RCG is set to consistent (which is the default mode), the data is applied at the destination only when the destination has a consistent image in the journal. When the mode is set to inconsistent, the data is applied at the destination on arrival without waiting for a full consistent image.

  Inconsistent mode is considered a risky state since a failure while in inconsistent mode may leave the target volumes with an inconsistent image. Inconsistent mode is used during a restore or reverse process after failing over. Inconsistent mode is useful when there has been a long network outage and it is likely that there will be insufficient room in the destination journal for the I/Os processed during the outage.

  You should consider creating a snapshot of the application volumes at the destination before initiating operations that may result in inconsistent mode.

- Pause mode – When the RCG is paused, all application I/Os are stored in the source journal. When replication of the RCG is resumed, the source SDR sends the journal contents to the target SDR to be applied to the target volumes. You may want to pause an RCG to handle a network issue between the peer systems or when fixing a hardware issue.
- Freeze mode – When the RCG is frozen, replication remains active. The application I/Os are replicated from the source SDR to the target journal but are not applied to the target volumes. When the RCG is unfrozen, the target SDR starts applying

the data in the target journal to the target volumes. You may want to freeze an RCG in order to create a snapshot of the replicated volume.

- Activity state - When the RCG is active, replication is enabled. When the RCG is inactive (either automatically inactivated by the system due to insufficient journal capacity, or the RCG is terminated manually by the user), replication is stopped, but RCG configuration is maintained. Activating an inactive RCG requires starting replication from the beginning with a full initialization.

Replication pairs

A replication pair is a pair of volumes with one volume at the source system and one at the target system. The source volume's data is replicated to the target volume.

The RCG can contain several replication pairs. A volume can be a member of at most one replication pair. When the replication pair is created, both the source volume and the target volume must be the same size. It is recommended, but not mandatory, that the volumes in the replication pair have the same attributes (including zero padding and granularity). Not doing so can impact performance and capacity.

Recovery point objective

The recovery point objective (RPO) is the maximum data loss, measured in units of time, that would be lost in a disaster or outage.

PowerFlex uses journal-based replication to achieve very low RPO and ensure minimal data loss. To ensure RPO compliance, PowerFlex replicates at least twice for every RPO period. For example, setting RPO to one minute means that PowerFlex can immediately return to operation at the target system with only one minute's worth of data loss. In order to achieve an RPO of one minute, replication takes place at least every 30 seconds. RPO compliance is calculated according to when the application data arrives at the target journal. Achieving RPO is the most important consideration in replication management. The user-defined RPO is the goal of replication. Even so, RPO compliance is not guaranteed. The system reports RPO compliance for each RCG.

Replication I/O flow

For replicated volumes, application data is processed by SDRs at the source and passed to the SDRs at the target.

In the source system, for replicated volumes, the SDCs communicate with the SDR. For non-replicated volumes, the SDCs communicate directly with the SDSs. Physical devices are connected to the SDS. The SDRs serve as a pipeline for I/Os. Application I/Os (both reads and writes) intended for replication volumes are sent from an SDC to an SDR. The source SDR packages the data into a consistent journal barrier and distills them so that only the most recent writes are included. The source SDR sends the journal barrier over the WAN to the target SDR. At the target system, the SDR processes the journal barrier and applies it to the volumes.

Replication initial copy

When replication is first activated for an RCG, the target volumes need to be synchronized with the source volumes.

For each replication pair, the entire contents of each source volume are copied to the corresponding target volume when replication is active. This occurs when the RCG containing the replication pair is activated, or when the replication pair is added to an already active RCG. When there is more than one replication pair in the RCG, the order in which the volumes are synchronized is determined by the order in which the replication pairs were activated. You can manually override this order by using the `--set_pair_initial_copy_queue_position` CLI command. You can also pause and resume the initial copy process using CLI commands. For detailed information on CLI commands, see the *Dell PowerFlex CLI Reference Guide*.

The initial synchronization occurs while all applications are running and performing I/O. Any writes to an area of the volume that has already been synchronized will be sent to the journal. Writes to an area of the volume that has not already been synchronized will be ignored, as the updated content will be copied over eventually as part of the synchronization.

The system has a limited number of RCGs. The system gives priority to replication pairs in the longest active RCGs. When an inactive RCG is activated, its replication pairs start the initial copy process. However, if there are insufficient system RCGs, initial copy priority will be given to replication pairs added to RCGs that are active longest. This means that replication pairs in a newly activated RCG might not start initial copy immediately.

Activate or terminate an RCG

Set the active/inactive state of a replication consistency group (RCG).

The default state of an RCG is active, though you can choose to create a new RCG in inactive state. When the RCG is active, replication is enabled.

When the RCG is in an inactive state, replication is stopped and the journal capacity is released. The I/Os are handled by the SDS and not the SDR and no journal capacity is consumed. All RCG configurations are maintained, including source and target roles, pair configuration, and volume access modes.

The RCG can be terminated automatically by the system when the journal capacity is fully consumed. You can also manually inactivate the RCG using the terminate command. When the RCG is terminated, volume I/Os are redirected from the SDR to the SDS. Peer connectivity is not required between the two sites. You can terminate the RCG at both sites separately. Alternatively, you can terminate the RCG at one site; once connectivity is restored, the peer will terminate the RCG at the other site.

Failover can be performed when the RCG is inactive, however consider that the data at the target may be old. While in failover mode, mappings at both the source and target can be activated and volumes can be written to, just like when RCG is in active state. Note that there is no data synchronization when the RCG is inactive. You can activate the RCG only once it is no longer in failover mode.

NOTE: Activating a terminated RCG requires full initialization. The initial copy process may take a long time.

The active/inactive RCG state and the commands to terminate or activate an RCG are only available if both peer systems are running PowerFlex v3.6.x or later.

Pause replication

Replication can be paused for a given RCG.

While paused, any application I/O is stored in the source journal. Once replication is resumed, the journal contents are sent to the target to be applied to the target volumes. You can pause replication when there are network issues, or you need to perform maintenance of troubleshooting on a hardware issue.

Freeze replication

You can freeze replication for a specific RCG.

While frozen, replication is still active and application I/O is passed from the source journals to the target journals. The target journals hold on to the data and do not apply them to the target volumes. You can freeze replication to take a consistent snapshot of the source or target volumes. It is important to note that leaving the RCG in a frozen state for a long period of time runs the risk of running out of journal capacity at the target system.

Using replicated volumes

Replicated volumes at the target are limited to read-only access.

While replication is active, the target volumes remain inaccessible to the hosts with only limited read-only access permitted. This is done to maintain data integrity and consistency between the two peer systems. The data at the target can be accessed to test replication, create snapshots, and to fail over.

Recovering from disaster

There are several ways to recover from a disaster. Doing so allows the target volumes to be accessible to the hosts and to take over as the replication source.

The following are options for recovering from a disaster:

- Failover—Fail over the RCG. During failover, the application I/Os are stopped at the source and the access mode of the source volumes is changed to inaccessible. The target volumes are brought to the newest consistent image available on the destination and their access mode is changed to read-write. Recovery from failover state can be achieved by either restoring the replication or reversing it.

- Test failover—Test failover of the RCG. This allows you to demonstrate failover mode without interrupting data synchronization.
- Switchover—Switch over the RCG. Switchover is a planned failover. The application I/Os are stopped at the source and the data is synchronized so that the target volume is consistent with the source volume. The access mode of the original source volumes is changed to unavailable or read-only. The access mode of the original target volumes is changed so that the hosts can read and write to the volumes.

Target volumes can be mapped only once failover or switchover has been performed. Once the RCG is in failover or switchover mode, you can decide how to continue with replication:

- Restore replication—This maintains the replication direction from the original source and overwrites all data at the target.
- Reverse replication—This changes the direction so that the original target becomes the source. All data at the original source is overwritten by the data at the target.

NOTE: Prior to executing the restore or reverse operation, you should consider creating a snapshot on the new target volumes in order to preserve the consistent image during the resynchronization process.

Replication direction and mapping

Use the table as a reference for replication direction and default access to volumes according to a subsequent RCG operation and action of the PowerFlex system.

The replication involves two peers, system A and system B. When the replication is set up, system A is set as source and system B is set as target. The following replication direction refers to the initial direction A->B and to changes to that direction.

Table 5. Replication direction and mapping according to subsequent RCG operations and possible actions:

Subsequent RCG operations	Possible actions	Replication direction/access	Access to volumes
Normal	Switchover/test failover/ failover Remove	A->B	Access to the volumes is allowed only through the source (system A)
After failover	Reverse/restore Remove	N/A - data is not replicated	By default access to the volume is allowed through the original target (system B). It is possible to enable access through the original source (system A).
After failover + reverse **NOTE:** Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	B->A	Access to the volumes is allowed only through the original target (system B)
After failover + restore **NOTE:** Switchover and test failover are only possible after the peers are synchronized.	Switchover/test failover/ failover Remove	A->B	Access to the volumes is allowed only through the source (system A)
After switchover	Switchover/test failover stop/failover Remove	B->A	Access to the volumes is allowed only through the original target (system B)
After test failover	Switchover/test failover/ failover Remove	A->B	Access to the volumes is allowed through both systems (system A and system B)

Setting up replication

In order to use replication in PowerFlex, you must first prepare the infrastructure.

The following is an overview of the necessary steps, which are detailed in the *Configure and Customize Dell PowerFlex*.

1. Add peer systems, at both the source and target systems. Add the corresponding certificates to each system.
2. Add journal capacity, ensuring sufficient capacity.
3. Add SDRs, at least two per system.

Once the replication infrastructure is set up, follow the following steps to configure replication:

1. Create Replication Consistency Group. Define RPO.
2. Add volume pairs. Volumes in a volume pair must have the same size. It is not necessary but is recommended that they have the same attributes (medium/fine granularity, zero padded/non-zero padded).
3. Start initial copy.

For detailed information about "PowerFlex: Introduction to Replication" under Manuals & Documentation, see *Managing PowerFlex Custom Node Using Dell OpenManage Enterprise* at http://www.dell.com/support

Protection and load balancing

PowerFlex maintains the user data in a RAID-1 mesh mirrored layout. Each piece of data is stored on two different Fault Units. The copies are distributed over the storage devices according to an algorithm that ensures uniform load of each fault unit by terms of capacity and expected network load. Rebuild and rebalance processes are fully automated, but are configurable.

Rebuild

PowerFlex initiates a Rebuild process in response to failure. Forward Rebuild refers to creating a new copy of the data on another server. Backward Rebuild refers to re-synchronizing one of the copies.

When a failure occurs, such as on a server, device or network failure, PowerFlex immediately initiates a process of protecting the data. This process is called Rebuild, and comes in two flavors:

- Forward rebuild is the process of creating another copy of the data on a new server. In this process, all the devices in the Storage Pool work together, in a many-to-many fashion, to create new copies of all the failed storage blocks. This method ensures an extremely fast rebuild.
- Backward rebuild is the process of re-synchronization of one of the copies. This is done by passing to the copy only changes made to the data while this copy was inaccessible. This process minimizes the amount of data transferred over the network during recovery.

PowerFlex automatically selects the type of rebuild to perform. This implies that in some cases, more data will be transferred to minimize the time that the user data is not fully protected.

Rebuild throttling

The rebuild throttling policy determines the priority of rebuild I/Os versus application I/Os when accessing SDS devices. The possible rebuild throttling policies are no limit on rebuild I/Os, limit concurrent I/Os per SDS device, Favor application I/Os and Dynamic Bandwidth Throttling.

Rebuild throttling sets the rebuild priority policy for a Storage Pool. The policy determines the priority between the rebuild I/O and the application I/O when accessing SDS devices. Please note that application I/Os are continuously served.

Applying rebuild throttling will on one hand increase the time the system is exposed with a single copy of some of data, but on the other hand, will reduce the impact on the application. One has to make a decision and choose the right balance between the two.

The following possible priority policies may be applied:

- No Limit: No limit on rebuild I/Os.

  Any rebuild I/O is submitted to the device immediately, without further queuing. Please note that rebuild I/Os are relatively large and hence setting this policy will speed up the rebuild, but will have the maximal effect on the application I/O.

- Limit Concurrent I/O: Limit the number of concurrent rebuild I/Os per SDS device (default).

The rebuild I/Os are limited to a predefined number of concurrent I/Os. Once the limit is reached, the next incoming rebuild I/O waits until the completion of a currently executed rebuild I/O. This will complete the Rebuild quickly for best reliability, however, there is a risk of host application impact.

- Favor Application I/O: Limit rebuild in both bandwidth and concurrent I/Os.

  The rebuild I/Os are limited both in bandwidth and in the amount of concurrent I/Os. As long as the number of concurrent rebuild I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebuild I/Os will be served. Once either threshold is reached, the rebuild I/Os wait until both I/O and bandwidth are below their thresholds. For example, setting the value to "1" will guarantee the device will only have one concurrent rebuild IO at any given moment, which will ensure the application IOs only wait for 1 rebuild IO at worst case.

  This imposes bandwidth on top of the Limit Concurrent I/Os option, which is a prerequisite to using this policy.

- Dynamic Bandwidth Throttling: This policy is similar to Favor Application I/O, but extends the interval in which application I/Os are considered to be flowing by defining a minimal quiet period. This quiet period is defined as a certain interval in which no application I/Os occurred. Note that the limits on the rebuild bandwidth and concurrent I/Os are still imposed.
- Default Values:
  - The default policy for rebuild is: Limit Concurrent I/O
  - Rebuild concurrent I/O Limit: 1 concurrent I/O

  NOTE:

  Rebuild throttling affects the system's performance and should only be used by advanced users.

Rebalance

When PowerFlex detects that user data is not balanced across devices in the Storage Pool, it initiates a process to restore the balance in which data copies are moved from the most utilized devices to the least utilized.

Rebalance is the process of moving one of the data copies to a different server. It occurs when PowerFlex detects that the user data is not evenly balanced across the fault units in a Storage Pool. This can occur as a result of several conditions such as: SDS addition/removal, device addition/removal, or following a recovery from a failure. PowerFlex will move copies of the data from the most utilized devices to the least utilized ones.

Both Rebuild and Rebalance compete with the application IO for the system resources. This includes network, CPU and disks. PowerFlex provides a very rich set of parameters that can control this resource consumption. While the system is factory-tuned for balancing between speedy rebuild/rebalance and minimization of the effect on the application IO, the user has very fine-grain control over the rebuild and rebalance behavior.

Rebalance throttling

The rebalance throttling policy determines the priority of rebalance I/Os versus application I/Os when accessing SDS devices. The possible rebuild throttling policies are no limit on rebalance I/Os, limit concurrent I/Os per SDS device, Favor application I/Os and Dynamic Bandwidth Throttling.

Rebalance throttling sets the rebalance priority policy for a Storage Pool. The policy determines the priority between the rebalance I/O and the application IO when accessing SDS devices. Please note that application I/Os are continuously served. Rebalance, unlike rebuild, does not impact the system's reliability and therefore reducing its impact is not risky.

NOTE:

Rebalance throttling affects the system's performance and should only be used by advanced users.

The following possible priority policies may be applied:

- No Limit: No limit on rebalance I/Os.

  Any rebalance I/O is submitted to the device immediately, without further queuing. Please note that rebalance I/Os are relatively large and hence setting this policy will speed up the rebalance, but will have the maximal effect on the application I/O.

- Limit Concurrent I/O: Limit the number of concurrent rebalance I/Os per SDS device.

  The rebalance I/Os are limited to a predefined number of concurrent I/Os. Once the limit is reached, the next incoming rebalance I/O waits until the completion of a currently executed rebalance I/O. For example, setting the value to "1" will

guarantee that the device will only have one rebalance IO at any given moment, which will ensure that the application IOs only wait for 1 rebalance IO in the worst case.

- Favor Application I/O: Limit rebalance in both bandwidth and concurrent I/Os.

  The rebalance I/Os are limited both in bandwidth and in the amount of concurrent I/Os. As long as the number of concurrent rebalance I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebalance I/Os will be served. Once either limiter is reached, the rebalance I/Os wait until such time that the limits are not met again.

  This imposes a bandwidth limit on top of the Limit Concurrent I/Os option.

- Dynamic Bandwidth Throttling: This policy is similar to Favor Application I/O, but extends the interval in which application I/Os are considered to be flowing by defining a minimal quiet period. This quiet period is defined as a certain interval in which no application I/Os occurred. Note that the limits on the rebalance bandwidth and concurrent I/Os are still imposed.
- Default Values:
  - The default policy for rebalance: Favor Application I/O
  - Rebalance concurrent I/O Limit: 1 concurrent I/O per SDS device
  - Rebalance bandwidth limit: 10240 KB/s

  For information on Instant Maintenance mode that uses rebuild and rebalance, see topic "Instant Maintenance mode".

Checksum protection

PowerFlex calculates and validates the checksum value for the payload during transit to protect data-in-flight. Checksum protection is applied to all I/Os.

In-flight checksum

This feature addresses errors that change the payload during the transit through PowerFlex. PowerFlex protects data in-flight by calculating and validating the checksum value for the payload at both ends.

NOTE: The feature is off by default for Medium Granularity.

NOTE: The checksum feature may have a significant impact on large block I/O latency. Contact Customer Support for more information.

- During write operations, the checksum is calculated when the SDC receives the write request from the application. This checksum is validated before each SDS writes the data on the storage device.
- During read operations, the checksum is calculated when the data is read from the SDS device, and the SDC validates the data before it returns to the application. If the validating end detects a discrepancy, it initiates a retry. The checksum is done in the granularity of a sector (1/2KB).

Persistent checksum

The persistent checksum feature applies to all I/Os—Application, Rebuild, Rebalance, and Migrate.

The checksum is also kept in RMcache (Read Memory Cache), protecting every block that is maintained in SDS memory against memory corruption. The checksum feature can be enabled at the Protection Domain level, and defined at the Storage Pool level. The feature is T10/DIF-ready.

- Pools with Fine Granularity with or without compression, have a persistent checksum by default. This checksum cannot be changed.
- Each I/O goes through compression, a checksum is calculated before it is written to the disk. There are two types of checksum:
  - Fine Granularity layout saves checksum data before and after processing to guarantee data integrity (compressed or not).
  - There are also system checksums for metadata.

Cache

PowerFlex offers a number of caching options, for the purpose of enhancing system performance.

The following caching options are supported by PowerFlex:

- RAM Read Cache (using DRAM server memory)
- Read Flash Cache (using SSD and NVMe SSD devices)

SSDs used for caching cannot be used for storage purposes.

NOTE: If CacheCade is used, you must configure External acceleration. Some devices are already benefitting from an external (For example, a non-PowerFlex related) acceleration/caching mechanism. This input is required in order to avoid alerts about media type mismatches. For example, when the user adds an HDD device which the SDS perceives as being too fast to fit the HDD criteria.

The following table summarizes information about the caching modes provided by the system.

Table 6. Caching modes

Mode	Description	Considerations	Default Setting
RAM Read Cache (RMcache)	Read-only caching performed by server RAM.	RAM Read cache, the fastest type of caching, uses RAM that is allocated for caching. Its size is limited to the amount of allocated RAM. **NOTE:** The amount that may be allocated is limited, and can never be the maximum available RAM.	Disabled, except when storage-only nodes are deployed.
Read Flash Cache (RFcache)	Read-only caching performed by one or more dedicated SSD or NVMe SSD devices in the server.	RFcache uses the full capacity of SSD devices (up to eight) to provide a larger footprint of read-only LRU (Least Recently Used) based-caching resources for the SDS. This type of caching reacts quickly to workload changes to speed up HDD Read performance. Several SSD devices can be allocated to a shared cache pool, and therefore the cache size is limited in size only by the amount of SSDs allocated for this purpose. The RFcache driver must be installed during deployment. Caching devices can be defined either during the installation process or after deployment. Limitations: The RFcache feature is only supported by 512 Byte devices for the cache use case. RFcache is not supported on 4K devices used for cache. However, RFcache acceleration of 4K storage devices is supported. Windows backend is not supported. Support matrix: • An RFcache device (flash device) can be partitioned only on Linux. • An SDS storage/source device cannot be partitioned if it needs to be accelerated by RFcache. • An SDS storage/source device as a file (over file system), cannot be accelerated by RFcache.	Disabled

Table 6. Caching modes (continued)

Mode	Description	Considerations	Default Setting
		• CloudLink Data at Rest Encryption (D@RE) is not supported on devices used for the RFcache feature.	

The following table illustrates the caching support matrix:

Table 7. Caching support matrix

System	RFcache	RMcache
PowerFlex	Yes	Yes
VxFlex Ready Node PowerEdge 13G servers	Yes	Yes
VxFlex Ready Node PowerEdge 14G servers	Yes	Yes

Networking

This section describes the various considerations for choosing how to manage inter-node communication - via a separate network with access to the PowerFlex components or on the same network.

In PowerFlex, inter-node communication (for the purposes of managing data locations, rebuild and rebalance, and for application access to stored data) can be done on one IP network, or on separate IP networks. Management (via any of the management interfaces) can be done in the following ways:

- Via a separate network with access to the other PowerFlex components
- On the same network

These options can be configured a) during deployment in the full PowerFlex Installer (via the CSV topology file) and using the VMware plug-in, as well as b) after deployment with the CLI.

The following table includes the list of PowerFlex networking requirements and supported architectures/configurations. For more detailed information, refer to the *PowerFlex Networking Best Practices and Design Considerations* document.

PowerFlex minimum network requirements and supported architectures	PowerFlex network limitations
• Ethernet based • Minimum supported speed: 2*10GbE • Recommended speed: 2*25GbE • Certified configurations: 4*25GbE, 2*100GbE, 4*100GbE • Supported MTU: 1500/9000 (jumbo frames recommended for optimal performance) • High availability and load balancing use at least two physical ports with: ○ PowerFlex native multi path across two or more subsets: ▪ Supports one to eight subnets ○ Link aggregation: ▪ A single IP subnet can be used if link aggregation is used, otherwise a minimum of two subnets is required. ○ It is recommended to use MDM virtual IP for simplified network stability in case of failures. • IP addresses: refer to guidelines about IP address calculation in IP addresses for R640. R740xd, and R840 servers • Data network latency requirements: ○ MDM: Maximum of 300 milliseconds • Switch redundancy:	• RMDA is not supported • Infiniband is not supported • IPv4 & IPv6 are supported on PowerFlex software only and VxFlex Ready Node offering • IPv4 is supported on PowerFlex appliance and PowerFlex rack offering • DHCP must not be deployed on a network where PowerFlex MDMs, SDS, SDR or SDCs reside • Firewall rules must allow PowerFlex ports - refer to PowerFlex Security guide for list of ports

PowerFlex minimum network requirements and supported architectures	PowerFlex network limitations
○ Switches must be redundant. This provides continued access to components inside the rack in the network incase a top of rack (ToR) switch fails. ○ Flat network is supported ○ Spine-leaf architecture is recommended for cross rack traffic ○ VXLAN is supported ● SDC connection via a router is supported and required for a low latency connection for optimal performance ● PowerFlex has a single management subnet which can be a dedicated port or a VLAN on a trunk connection. For best practices, refer to: ○ VxFlex Ready Node R640-R740xd-R840 Operating System Installation and Configuration Guide for Linux ○ VxFlex Ready NodeR640-R740xd-R840 Operating System Installation and Configuration Guide for ESXi	

PowerFlex network requirements and supported architectures for systems with replication	PowerFlex Network limits for systems with replication
● Minimum best practice with replication: 4*25GbE or 2*100GbE ● Additional IP addresses for site to site routing includes: ○ Within a protection domain, SDRs are installed on the same hosts as SDSs. The traffic that an SDR writes to the journal volume, is sent to all SDSs that host the journal, not only the one that is co-located on the host. In the backend storage network, each SDR listens on the same node IPs as the SDSs and should be able to reach all SDSs within the protection domain. ○ The SDRs require additional, distinct IP addresses which allow them to communicate with remote SDRs. In most cases, these should be routable addresses with a properly configured gateway. For redundancy, each SDR should have two. ● Static routes: ○ PowerFlex asynchronous replication usually happens over a WAN between physically remote clusters that do not share the same address segments. If the default route itself is not suitable to properly direct packets to the remote SDR IPs, static routes must be configured to indicate either the next top address or the egress interface or both for reaching the remote subnet. ○ PowerFlex Manager can add static routes for replication use cases	● MDM to MDM peer metadata synchronization should take place over a WAN with less than 200ms latency ● Local SDR to remote SDR: Latency is not as sensitive in SDR > SDR traffic, but round-trip time should not be greater than 200ms ● Firewall rules must allow PowerFlex ports - refer to PowerFlex Security guide for list of ports used by replication

Refer to the following configuration guidelines used by VxFlex Ready Node for:

- Linux examples: R640 and R740xd nodes refer to Configuring network ports on Linux servers
- ESXi examples: R640, R740xd, and R840 nodes refer to Configuring network ports on ESXi hosts

This section describes how to choose from these options, depending on your organization's requirements, security considerations, performance needs, and IT environment.

PowerFlex networking considerations:

- Single IP network: All communications and IOs used for management and for data storage are performed on the same IP network. This setup offers the following benefits:
  - Ease of use
  - Fewer IP addresses required

- Multiple separate IP networks: Separate networks are used for management and for data storage, or separate networks are used within the data storage part of the system. This setup offers the following benefits:
  - Security
  - Redundancy
  - Performance
  - Separate IP roles in order to separate between customer data and internal management

  NOTE: Network high availability can be implemented by using NIC-bonding (refer to https://www.dell.com/en-us/dt/storage/powerflex.htm#tab0=1&pdf-overlay=//www.delltechnologies.com/asset/en-us/products/storage/industry-market/powerflex-white-paper-networking-best-practices.pdf) or by using several data networks in PowerFlex.

  For more information about MTU performance considerations and best practices, see the *PowerFlex Configure and Customize Guide*.

  NOTE: The MDM cluster IP address limit is 16 IP addresses, which includes all cluster members (primary, secondary and standby).

  The following table describes the range of potential IP address configurations:

Table 8. IP address configurations in PowerFlex (based on CSV file)

Column in CSV file	MDM Mgmt IP	MDM IPs	SDS All IPs	SDS-SDS Only IPs	SDS-SDC Only IPs
Comments	Management Access	Control Network	Rebuild and Data Path Network	Rebuild Network	Data Path Network
	Optional, but recommended; not applicable for tiebreaker IP addresses that can be used to provide access to PowerFlex management applications, such as PowerFlex GUI, CLI, REST API, OpenStack. This IP address must be externally accessible.	Mandatory IP addresses used for MDM control communications with SDSs and SDCs, used to convey data migration decisions, but no user data passes through the MDM. Must be on the same network as the data network. Must be externally accessible if no MDM Management IP addresses are used. MDM Virtual IP is supported on the data network.	IP addresses used for both SDS-SDS and SDS-SDC communications. These IP addresses will also be used to communicate with the MDM	IP addresses used for SDS-SDS communication only. These addresses are used for rebuild & rebalance operations. These IP addresses will also be used to communicate with the MDM.	IP addresses used for SDS-SDC communication. These addresses are only used for read-write user data operations.

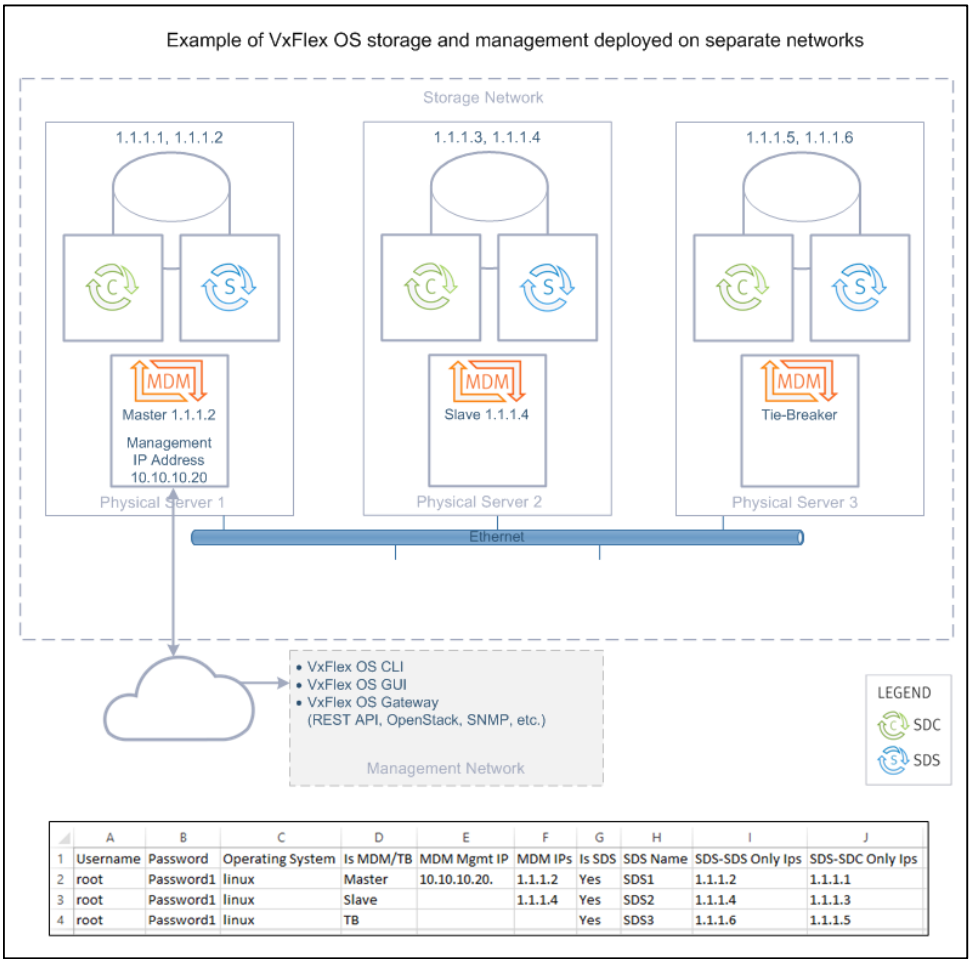
  The following combinations can be used for SDS/SDC:

- Only *SDS All IPs*
- Only *SDS-SDS Only IPs* + *SDS-SDC Only IPs*
- *SDS All IPs* + either *SDS-SDS Only IPs* or *SDS-SDC Only IPs* (can be used in cases of multiple networks; ensure that you do not use the same IP address more than once in the networks).
- *SDS All IPs* + both *SDS-SDS Only IPs* and *SDS-SDC Only IPs* (can be used in cases of multiple networks; ensure that you do not use the same IP address more than once in the networks).

NOTE:

In the following example drawing for separate networks, a very simple example is shown, where the management and storage parts of the system are on different networks. In more complex configurations, MDMs, SDCs and SDSs can be on separate networks. Up to 8 separate networks per PowerFlex installation are supported.

The following figure shows an example configuration and the corresponding fields in a CSV configuration file:



	A	B	C	D	E	F	G	H	I	J
1	Username	Password	Operating System	Is MDM/TB	MDM Mgmt IP	MDM IPs	Is SDS	SDS Name	SDS-SDS Only Ips	SDS-SDC Only Ips
2	root	Password1	linux	Master	10.10.10.20.	1.1.1.2	Yes	SDS1	1.1.1.2	1.1.1.1
3	root	Password1	linux	Slave		1.1.1.4	Yes	SDS2	1.1.1.4	1.1.1.3
4	root	Password1	linux	TB			Yes	SDS3	1.1.1.6	1.1.1.5

Figure 2. PowerFlex deployed on separate networks

IPv4 and IPv6

IPv6 is supported only on the PowerFlex software-only and VxFlex Ready Node offerings. The PowerFlex appliance and PowerFlex rack offerings currently do not support IPv6.

VMware limitation

When deployed on VMware, PowerFlex, the same IP Subnet can be configured on different VM kernel interfaces\virtual switches.

For more information, see the VMware limitation in the following link:

https://kb.vmware.com/s/article/2010877?lang=en_us

PowerFlex only supports the following network configurations when deployed on VMware:

- A single data storage network
- Two or more data networks, each on separate IP subnets
- A single IP data network using several NIC-bonding configurations, or vSwitch load balancing

Network stability

Application level multipathing configurations:

Resilient network connectivity is standard for PowerFlex. Nevertheless the actual implementation many vary. In some cases, the resiliency is attained through LACP, while other cases through application level multipathing.

In PowerFlex v3.6, additional capabilities were added to support the following error scenario use cases:

Use case one:

1. Partial failures on existing connections: While many network issues are manifested as a complete disconnection, some does not, PowerFlex is now aware of degraded connections. Types of degraded connections are:
   a. Stable connection and yet very slow vs others.
   b. Unstable connection (occasional disconnections)
   c. Combination of the above two

Use case two:

1. SDC to SDS Proxy connection (in the PowerFlex architecture, each SDC is required to be fully connected to all the SDSs on each of the PDs it consumes volumes from. In case the SDC is unable to reach one or more of the SDSs, the MDM will reflect that and the management API's will raise an alert. In PowerFlex v3.6 a capability was introduced:
   a. In case the SDC cannot access the SDS it will send the IO to another SDS within the same PD (or subset of this PD). We will refer to this SDS as Proxy-SDS since its only role is to forward the IO to the SDS which owns the data, receive the IO result and return it to the initiating SDC.

Storage data client host

PowerFlex supports the storage data clients (SDCs) host to map and present volumes to the operating system.

The SDC is a PowerFlex-proprietary lightweight device driver that provides frontend volume access to applications, and maps and presents volumes to the operating system or hypervisor. SDCs maintain peer-to-peer connections to every SDS managing a pool of storage. They translate SCSI to PowerFlex protocol.

Storage data clients

The storage data client is a driver installed on hosts that translates the PowerFlex control and data path protocols to the SCSI layer. The SDC handles all the PowerFlex multipathing aspects and connects to all the PowerFlex storage components.

The SDC also integrates with relevant operating system logic to provide a native block storage device in the operating system. Volumes (also know as LUNs in other storage systems) are mapped to the SDC to be consumed as block devices.

SDCs are supported on Linux, Windows, AIX, and other operating systems, and can be installed on any non-PowerFlex host meeting PowerFlex specifications for SDCs. PowerFlex appliance and PowerFlex rack also support installing SDCs in hyperconverged configurations.

As shown in the following figure, the SDC connects directly to the storage layer.

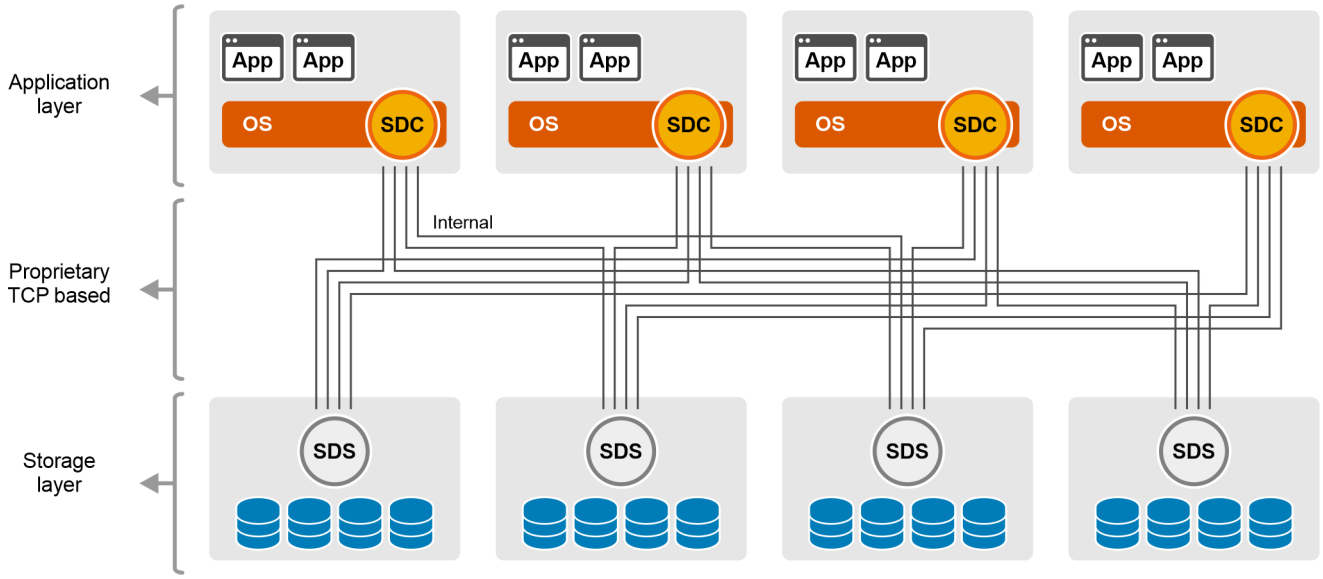


Figure 3. SDC host connectivity

SCSI support

The SCSI identity consists of the following:

- The ESXi SDC registers a new SCSI adapter with Fibre Channel transport.
- The Fibre Channel node worldwide name is the low 64 bits of the SDC GUID.
- The Fibre Channel port worldwide name is the high 64 bits of the SDC GUID.
- The Fibre Channel port and node worldwide names are also exposed for the virtual Fibre Channel targets.
- Every mapped volume contains the following:
  - A SCSI worldwide name which can be queried at a vital product data (VPD) page `0x83` EUI-64 designator using the SCSI Inquiry command. The EUI-64 designator is 128 bits, and it is composed of the PowerFlex system MDM ID and Volume ID.
  - A SCSI unit serial number can be queried though a VPD page `0x83` EUI-64 designator using the SCSI Inquiry command. The serial number is made of `EMC - MDM_ID-Vol_ID`
  - **`MOS_SNPRINTF((char *)&pUnitSerial->serialNum[0], serialNumBufSize, "%s-%.16"MOS_X64_BIT__STRING"-%.16"MOS_X64_BIT__STRING, MOS_VERSION__LEGACY_COMPANY_NAME_CAP, mapVol_GetMdmID(pVol), mapVol_GetVolID(pVol));`**
- Every mapped volume is assigned a Target LUN address when it is mapped. Because the storage I/O (SIO) volume ID is 64 bits, the address is calculated as follows:
  - The SDC exposes up to 8 K LUNs in total. Begin with the volume index (low 32 bits of the volume ID) and use the modulo operation with 8 K. The result is called the SCSI volume index.
  - The low eight bits of the SCSI volume index become the LUN ID, and the remaining top five bits becomes the target ID. If there is a collision, which is when a volume exists with a TargetLun pair, the LUN ID stays a low eight bits and the target ID increments by one until the collision is resolved. If a collision exists for all 32 target IDs, the LUN ID increments by one and starts over until the collision is resolved.
  - SDC reports a single target worldwide name that is based on the SDC GUID. This target ID is unrelated to the target that is part of the volume LUN ID.

Supported SCSI commands

The following tables describe the SCSI commands that PowerFlex supports.

Table 9. Mandatory SCSI commands

Command type	Command code		Comments
Format Unit			--I
Inquiry	#define SCSI_CODE__INQUIRY	0x12	INQUIRY commands support a subset of the standard pages.

Table 9. Mandatory SCSI commands (continued)

Command type	Command code		Comments
Read	#define SCSI_CODE__READ10	0x28	--
Read	#define SCSI_CODE__READ6	0x8	--
Read	#define SCSI_CODE__READ12	0xA8	--
Read	#define SCSI_CODE__READ16	0x88	--
Read Capacity	#define SCSI_CODE__READCAPAC10	0x25	--
Read Capacity	#define SCSI_CODE__READCAPAC16	0x9E	--
Report LUNs	#define SCSI_CODE__REPORTLUNS	0xA0	--
Test Unit Ready (TUR)	#define SCSI_CODE__TEST_UNIT_READY	0x00	--

Table 10. Optional SCSI commands

Command type	Command code		Comments
Compare and Write	#define SCSI_CODE__COMPARE_AND_WRITE	0x89	Atomic test and set (ATS)
Mode Sense - only page `0xa` /subpage `0x0`	#define SCSI_CODE__MODE_SENSE	0x1A	MODE_SENSE commands support a subset of the standard pages.
Reservations related			
Persistent Reserve In (SCSI-3)	#define SCSI_CODE__PR_IN	0x5E	--
Persistent Reserve Out (SCSI-3)	#define SCSI_CODE__PR_OUT	0x5F	--
Reserve (SCSI-2)	#define SCSI_CODE__RESERVE6	0x16	--
Reserve (SCSI-2)	#define SCSI_CODE__RESERVE10	0x56	--
Release (SCSI-2)	#define SCSI_CODE__RELEASE6	0x17	--
Release (SCSI-2)	#define SCSI_CODE__RELEASE10	0x57	--
Unmap	#define SCSI_CODE__UNMAP	0x42	Thin provisioning or block delete
Write	#define SCSI_CODE__WRITE10	0x2A	--

Table 10. Optional SCSI commands (continued)

Command type	Command code		Comments
Write	#define SCSI_CODE__WRITE6	0x0A	--
Write	#define SCSI_CODE__WRITE12	0xAA	--
Write	#define SCSI_CODE__WRITE16	0x8A	--
Write Same	#define SCSI_CODE__WRITE_SAME10	0x41	Zero block or write zeroes
Write Same	#define SCSI_CODE__WRITE_SAME16	0x93	Zero block or write zeroes
Write Same	#define SCSI_CODE__WRITE_SAME32	0x7F	Zero block or write zeroes

The following command is unsupported:

- Extended Copy

Supported VPD pages

PowerFlex supports the following pages for SCSI Vital Product Data (VPD):

- `0x00` - Standard inquiry
- `0x80` - Unit serial number
- `0x83` - Device identification
- `0x80` - Block limits
  - This page includes size limitations used by the optional commands: Unmap, Write Same, and Compare and Write.
- `0x83` - Logical block provisioning

Monitor of SDC connections

The system monitors all connections between SDCs and SDSs and sends out an alert when an active connection between an SDC and an SDS goes down.

To effectively monitor SDC and SDS connections, the MDM collects connectivity updates from all of the SDCs. The MDM posts events whenever an SDC connects to or disconnects from a specific SDS IP address. The MDM frequently analyzes the connectivity status to determine the current system state. The system does not send out alerts for temporary connectivity issues that are resolved in less than 10 seconds.

The following are the possible connectivity states between SDCs and SDSs in the system:

- All connected
- One SDC is disconnected from one SDS
- One SDC is disconnected from one SDS IP address
- One SDC is disconnected from all SDSs
- All SDCs are disconnected from one SDS
- All SDCs are disconnected from one SDS IP address
- All SDCs are disconnected from all SDSs
- Multiple disconnections

When the system's connectivity state changes to any state other than `All Connected`, an alert is displayed in the PowerFlex GUI and is written to the MDM event log. Once an alert is generated, you can use the SCLI to query details on the disconnection using the command `scli --query_sdc_to_sds_disconnections`. For more information about running SCLI commands, see the *CLI Reference Guide*.

The MDM does not monitor the connectivity state of SDCs or SDSs in the following scenarios:

- SDS is in maintenance mode
- SDS is disconnected from the MDM
- SDS is in the process of being removed
- SDC is disconnected from the MDM for more than two minutes
- SDC is not approved

The alerts are detailed in "Alerts in SNMP, PowerFlex GUI, REST, and ESRS" in the *Monitoring PowerFlex* .

S.M.A.R.T. hardware monitoring

The PowerFlex bare-metal solution now provides monitoring capabilities for RAID controllers and storage devices compatible with S.M.A.R.T. (Self-Monitoring, Analysis and Reporting Technology) protocols.

In Linux-based environments, S.M.A.R.T.-compatible HDDs, SSDs and RAID storage controllers can be monitored for S.M.A.R.T. attributes such as temperature, SSD wear level, and error counters. LEDs can also be lit on these hardware devices, to simplify physical identification for maintenance purposes.

Each hardware vendor defines specific thresholds for the S.M.A.R.T. attributes. This feature currently supports storage devices controlled by LSI, HP and Dell RAID controllers, and stand-alone devices. During system deployment, an external monitoring tool is installed as part of the LIA on each node. Additional RAID controller tools must be installed manually after system deployment: storcli for LSI RAID controllers, hpssacli for HP RAID controllers, or perccli for DELL RAID controllers. These tools are used by the system to collect the counters that are returned to the MDM.

NOTE: In some cases, LSI RAID controllers may report vendor information as "AVAGO" instead of LSI.

The MDM queries the SDSs at set intervals, and stores the returned information. This information can be viewed using CLI queries. In addition, when thresholds are crossed for S.M.A.R.T. attributes, alerts are generated by the system.

When the CLI is used to query device information, physical device information, such as serial number, disk slot information, model name, vendor etc., temperature, and wear level information (for SSDs only) is included in the returned response.

For information about the use of CLI commands, see the *PowerFlex CLI Reference Guide*.

For information about the use of REST API URIs, see the *PowerFlex User Guide*.

You can use the PowerFlex GUI to monitor S.M.A.R.T.-related alerts in the **Alerts** view.

In addition, SNMP traps and ESRS alert codes can be used to monitor alerts triggered by devices compatible with S.M.A.R.T.

List of approved RAID controllers

Provides high-level specifications of RAID controllers, which are tested and certified by PowerFlex.

PowerFlex-certified RAID controllers

The following table describes the PowerFlex-certified RAID controllers:

Manufacturer	Specifications
HP	• Model Name: Smart Array P440ar • Vendor Name: HP • Firmware Version: 3.56 • Driver Version: 3.4.10 • Driver Name: hpsa
DELL	• Model Name: PERC H730 Mini • Vendor Name: Dell • Firmware Version: 25.5.5.0005 • Driver Version: 07.700.52.00 and above • Driver Name: megaraid_sas

Manufacturer	Specifications
	• Model Name: H740p • Vendor Name: Dell • Firmware Version: Refer to VxFlex Ready Node Firmware and Driver support matrix • Driver Version: Refer to VxFlex Ready Node Firmware and Driver support matrix • Driver Name: megaraid_sas
	• Model Name: HBA330 • Vendor Name: Dell • Firmware Version: Refer to VxFlex Ready Node Firmware and Driver support matrix • Driver Version: Refer to VxFlex Ready Node Firmware and Driver support matrix • Driver Name: mpt3sas

NVDIMM hardware awareness feature

From PowerFlex3.6-<build>, the customre can enalbe the NVDIMM hardware awareness feature on the SDS.

When this feature is enabled, the SDS detects bad blocks on NVDIMM devices used for FG acceleration and takes appropriate actions based on how and when the corruptions in the NVDIMM devices are detected. Refer to the *Upgrade PowerFlex*, *Deploy PowerFlex* and *Configure and Customize PowerFlex* guides for the procedure on how to configure.

Snapshots

PowerFlex enables you to create instantaneous copies of volumes, i.e. snapshots. A snapshot can be manipulated as any volume exposed by the storage system. Snapshots taken at the same time form a consistency group and can be manipulated as a group.

The PowerFlex storage system enables you to take snapshots of existing volumes, up to 128 per volume. The snapshots are thin provisioned and are extremely quick. For more information about thin provisioning, see "SAN virtualization layer".

Once a snapshot is generated, it becomes a new, unmapped volume in the system. You can manipulate it in the same manner as any other volume exposed by the PowerFlex storage system.

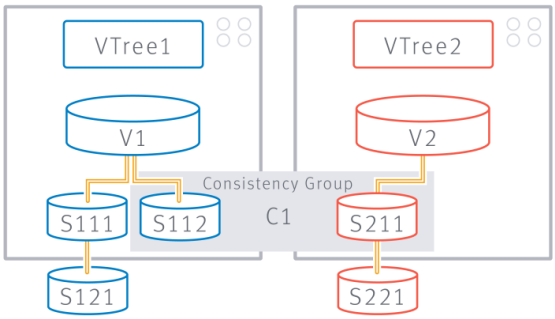


Figure 4. Snapshot operations

The structure related to all the snapshots resulting from one volume is referred to as a V-Tree (short for volume tree). When taking a snapshot in the system, you can specify more than one volume. All snapshots taken together form a consistency group. They are consistent in the sense that they were all taken at the same time. So if there is a contextual relationship between the data contained by all the snapshot members, then that set is meaningful. The consistency group allows manipulation of the entire set.

If you remove an entire consistency group, all of the snapshots that were taken together will be removed. In RED, S211 is a snapshot of V2. Since S112 and S211 were taken together, they compose a consistency group designated as C1.

NOTE: The consistency group is only for convenience purposes. There are no protection measures done by PowerFlex to conserve the consistency group. For example, you can remove a snapshot that is a member of a consistency group.

NOTE: Support of removal of a parent snapshot in the system. You can merge the parent to a child snapshot.

NOTE: It is recommended not to trim volumes that contain snapshots as the logical capacity may not shrink.

Volume trees (V-Trees)

A V-Tree consists of the root volume and all of its descendant volumes and snapshots.

A V-Tree is the structure comprised of a volume and the snapshots resulting from that volume. It is a tree spanning from the source volume at its root, whose descendants are either snapshots of the volume itself or snapshots of a snapshot. In the V-Tree diagram, S_{111} and S_{112} are snapshots of V_1. S_{121} is a snapshot of snapshot S_{111}. Together, V_1 and S_{1xy} are the V-Tree of V_1.

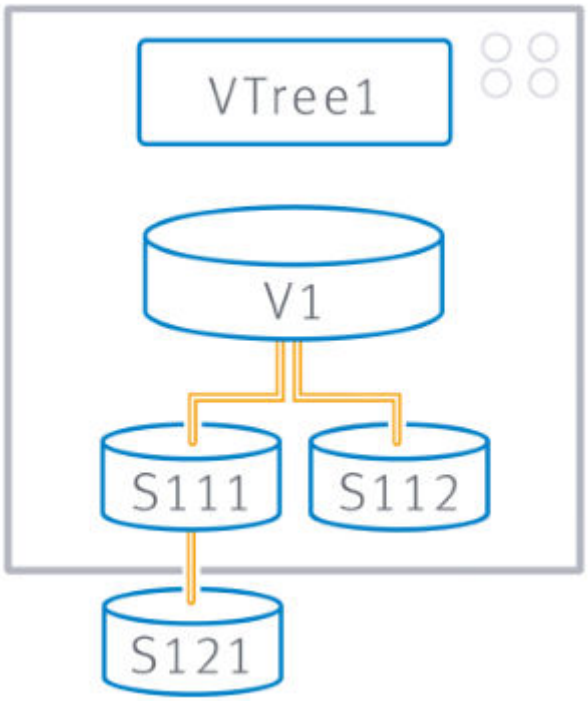


Figure 5. V-Tree diagram

Snapshot policies

Snapshot policies enable you to define policies where you can configure the number of snapshots to take at a given time for one or more volumes.

Snapshots are taken according to the rule defined. You can define the time interval in-between two rounds of snapshots as well as the number of snapshots to retain, in a multi-level structure. For example, take snapshots every x minutes/hours/days/ weeks. There are one to six levels, with the first level having the most frequent snapshots.

Example:

Rule: Take snapshots every 60 minutes

Retention Levels:

- 24 snapshots
- 7 snapshots
- 4 snapshots

After defining the parameters, select the source volume to add to the snapshot policy. You can add multiple source volumes to a snapshot policy, but only a single policy per source volume is allowed. Only one volume per vTree may be used as a source volume of a policy (any policy). For more information on vTree, see "Snapshots" in the *Getting to Know Guide*.

When you remove the source volume from the policy, you must choose how to handle snapshots. Snapshots created by the policy are referred to as auto snapshots. In the PowerFlex GUI, this is indicated if a snapshot policy is displayed for the snapshot.

- If the source volume has no auto snapshots, you can't unassign the source volume from the snapshot policy. You can remove auto snapshots from **Snapshots** in the PowerFlex GUI.
- If the source volume has auto snapshots but none of them are locked, you are prompted to confirm that you would like to delete all auto snapshots. If any of the auto snapshots are locked, the locked auto snapshots are just detached from the snapshot policy, but not deleted. You need to manually remove auto snapshots. It doesn't matter if the auto snapshot is locked or not, you can still delete.

Performance

System performance can be enhanced by implementing the high performance profile. This is the default profile for all system objects (MDMs, SDSs, SDCs, and SPs).

The profile of an SDS can't be changed to compact if it has at least on Fine Granularity device

The MDM, SDC, and SDS components are deployed with the default performance profile. To enhance system performance, you can configure each component with the high profile, either during or after deployment.

The high performance profile should be enabled on servers equipped with processors based on Haswell-equivalent or newer architectures and 48 GB of RAM. Other configurations may benefit from this setting as well.

To configure the performance profile during deployment, see the "Configure performance profile during deployment section" of the *PowerFlex Deployment Guide*.

To configure the performance after deployment, use the PowerFlex GUI, as described in the *PowerFlex Configure and Customize Guide User Guide*. You can also use the `set_performance_parameters` command, as described in the *PowerFlex CLI Reference Guide*.

For more information on the difference between the default and the high performance profile, see the *PowerFlex Configure and Customize Guide* .

Migrate volumes

You can migrate volumes to a different Storage Pool or Protection Domain within your PowerFlex system.

You can migrate volumes from the following storage layers:
- From one Storage Pool to another within the same Protection Domain
- From one Storage Pool to another across different Protection Domains
- Volume Tree and all it's snapshots are migrated together
- From thick to thin
- From Fine/Medium to Medium/Fine Granularity

Requirements to migrate volumes:
- Medium Granularity pools should have zero padding enabled

Implementing PowerFlex

This section outlines the steps for implementing PowerFlex.

Implementing PowerFlex is, in general, a two-step process: first build the physical storage layer, then configure the virtual SAN layer on top of it.

Communication is done over the existing LAN data network using standard TCP/IP. The MDM and SDS nodes can be assigned up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

To build the physical storage layer:
1. Physically assemble the server nodes in the rack and perform all cabling, according to the instructions for your PowerFlex product.
2. Configure the operating system, including all required IP addresses, based on network best practice guidelines.
3. Using PowerFlex Installer, deploy the PowerFlex software.

The physical layer is now ready, and you can expose a virtual storage layer.

PowerFlex appliance and PowerFlex rack offer a more automated installation experience using PowerFlex Installer. For more information regarding PowerFlex Manager management and operation capabilities, refer to PowerFlex appliance and PowerFlex rack product information.

PowerFlex implementation in an ESXi-based system

In the VMware environment, the MDM and SDS components are installed on a dedicated SVM, and the SDC is installed directly on the ESXi host.

Minimum requirements

These are the minimum system requirements for a PowerFlex implementation in an ESXi-based system:

- Three ESXi servers with 240 GB of free capacity per server
- 10 Gbps network

ESXi 6.0	ScaleIO v2.0.1.4 and v2.5
	VxFlex OS v2.6.x and v3.0.x
ESXi 6.5	ScaleIO v2.0.1.4 and v2.5
	VxFlex OS v2.6.x and v3.0.x
	PowerFlex v3.5 and v3.5.1[1, 2]
ESXi 6.7	VxFlex OS v3.0.x
	PowerFlex v3.5 and v3.5.1 [1, 2]
ESXi 7.0	PowerFlex v3.5 and v3.5.1 [1, 2]

1. Asynchronous replication available with PowerFlex v3.5 is limited to a 2-layer configuration at this time.
2. Asynchronous replication is available with PowerFlex 3.6 as part of the PowerFlex appliance and Integrated Rack offerings.
3. Currently upgrade from ESXi 6.7 to ESXi 7.0 with PowerFlex v3.5, v3.5.1 requires SDC removal and re-installation. For detailed upgrade procedure please refer to PowerFlex Upgrade Guide.

Components

The PowerFlex hyper-converged configuration consists of a dedicated Storage Virtual Machine (SVM) and three software components:

- Storage Virtual Machine (SVM): A Linux-based virtual machine dedicated to PowerFlex and is used to host the different PowerFlex software components described here.
- Meta Data Manager (MDM): Configures and monitors the PowerFlex system. The MDM is configured in a redundant cluster mode. The MDM is installed on the SVM. The MDM can be configured as a three-node cluster (primary MDM, slave MDM and tiebreaker MDM) or as a five-node cluster (primary MDM, 2 secondary MDMs and 2 tiebreaker MDMs) to provide greater resiliency.
- Storage Data Server (SDS): Manages the capacity of a single server and acts as a back-end for data access. The SDS is installed on all servers contributing storage devices to the PowerFlex system. These devices are accessed using SDS. The SDS is installed on the SVM.
- Storage Data Replicator (SDR): Handles volumes replication activities between PowerFlex systems through journal management. The SDR is located alongside the SDS.
- Storage Data Client (SDC): A lightweight device driver that presents PowerFlex volumes as block devices to the application located on the server on which the SDC is installed. The SDC creates a logical adapter, which is an ESXi kernel construct. The adapter informs ESXi about the arrival and disappearance of SCSI devices. These LUNs can be formatted with VMFS and then exposed using the ESXi host to the virtual machine or can be used as RDM devices. This is depicted in PowerFlex implementation in ESXi with SDC in VMkernel. Note that the SDC is installed within the ESXi kernel similar to any other VIB.

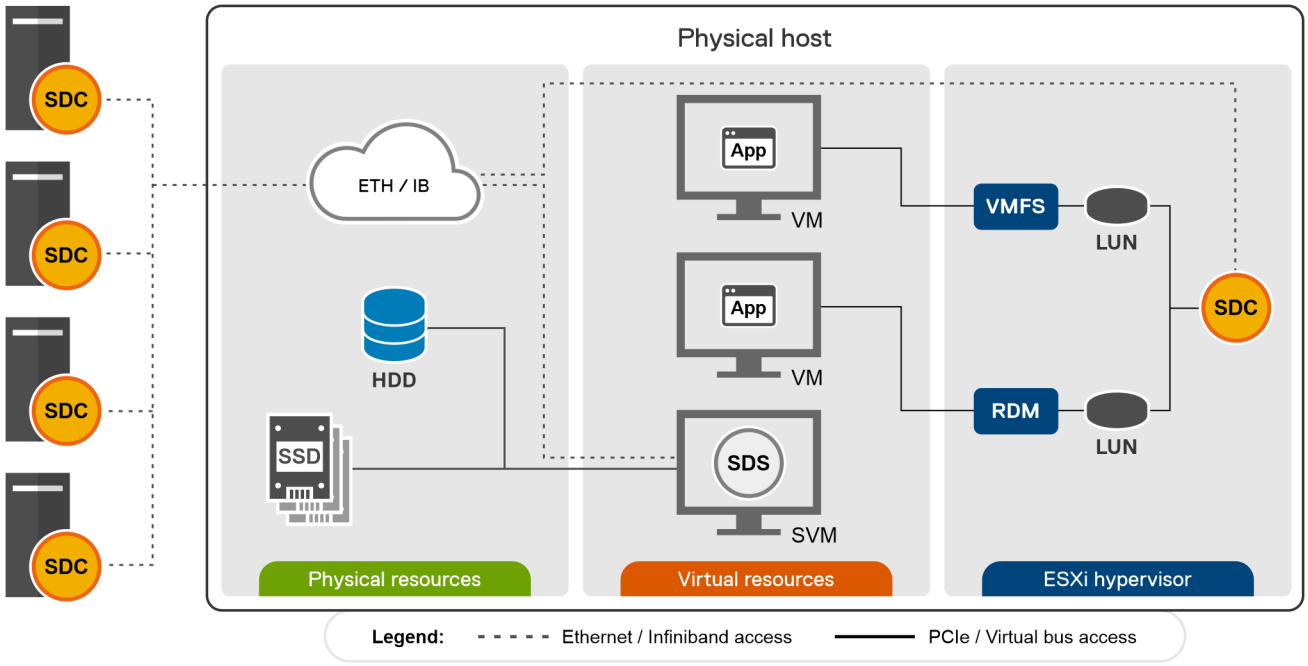


Figure 6. PowerFlex implementation in ESXi with SDC in VMkernel

The LUNs in this example can be formatted with VMFS, and then exposed using the ESXi host to the virtual machine, or can be used as RDM devices. When the LUNs are used as RDM devices, the VMFS layer is omitted.

Supported Features

The following VMware features are supported by PowerFlex:

- vMotion
- Storage vMotion
- Fault Tolerance
- DRS
- Storage DRS
- VAAI (except Full Copy primitive)
- Virtual Volumes (vVols)

VMware APIs for Storage Awareness (VASA)

PowerFlex (v3.0.1 and later) supports VMware vVols 1.0 through VASA 2.0 implementation.

- A PowerFlex cluster may contain a single VASA provider or three for a highly resilient configuration using replica sets.
- The PowerFlex SDC on each host exposes a Protocol Endpoint (PE) for each VASA provider.

PowerFlex VASA implementation exposes four profiles that can be used to implement VM storage policies, based on the underlying storage configuration or capabilities (SPBM):

- Bronze—HDD-based Storage Pool
- Silver— Accelerated HDD storage Pool (RFcache)
- Gold—All SSD Storage Pool
- Compression—Fine Granularity (FG) Storage Pool (SSD-based)

Device management

Devices can be managed in the following ways:

- VMDirectPath I/O

Device management is performed via the SVM, yielding the best I/O performance. Devices are added to the system after the deployment. On factory-installed VxFlex Ready Node servers, the vSphere plug-in configures DirectPath on the ESXi servers and adds devices to the system. On other servers, you must enable VMDirectPath manually to the ESXi server, then use the plug-in to add devices.

Requirements:

- When DirectPath is configured, all servers in the system must use DirectPath. For VxFlex Ready Node systems, this can be configured with the PowerFlex vSphere plug-in. For other servers, use the vSphere client to configure each ESXi host manually.
- If the host has multiple controllers, you must configure DirectPath on that host manually, not with the plug-in. After that is done, you can use the plug-in to deploy PowerFlex on that host and to add devices to it.
- All devices on all servers must not have any other use (not VMDK, RDM, or be part of a datastore).
- PowerEdge 14G servers with any NVME devices cannot be added in a DirectPath-based system. Use RDM-based, instead.
- ESXi boot device requirements:
  - Must be on a separate controller or connected directly to the board.
  - SATADOM and M2 boot devices are supported. These devices allow the creation of a datastore on the system disk, which is needed to host the SVM.
  - USB boot devices are not supported.

NOTE: DirectPath architecture is the recommended best practice for RAID and SAS Controller managed drives.

- RDM

  Using RDM mapping, a device is created on the SVM that points to the physical disk on the ESXi.

  You can add RDM devices that are connected through a physical RAID controller. If a local RDM is not connected via a RAID controller, it may not be supported. To ensure the compatibility of these devices, you can add them as VMDK, or you can select **Force RDM on device with unknown RDM support**, as described in the "Advanced settings options" section of the *PowerFlex Deployment Guide*. Enable this option before beginning the deployment.

  Before enabling this feature, contact Customer Support.

- VMDK

  A new datastore is created, with a VMDK, and the VMDK is added to the SVM. PowerFlex requires thick provisioning, so this process can take a long time.

  In almost all cases, RDM is the preferred method to add physical devices. Use the VMDK method only in the following scenarios:

  - The physical device does not support RDM.
  - The device already has a datastore, and the device isn't being completely used. The excess area that is not already being used will be added as a PowerFlex device.

  NOTE: To use VMDK, select **Enable VMDK creation**, as described in the "Advanced settings options" section of the *PowerFlex Deployment Guide*.

System size

If you are deploying a very large PowerFlex system (hundreds of nodes), you can increase the default parallelism limit (default: 100), thus speeding up the deployment. This is dependent on the processing power of the vCenter.

To increase the parallelism limit, use the plug-in **Advanced settings**, as described in the "Advanced settings options" section of the *PowerFlex Deployment Guide*.

Pre-deployment considerations

You should take these considerations into account before deploying the system:

- Do you want to use separate networks for data and management (recommended), and which IP addresses will you use for the SVMs and VMkernels?
- Are there flash devices that will be added to the SDS?

- Do you want to create Fault Sets? See the requirements in "Fault Sets".

Post-deployment considerations

You should take these considerations into account after deploying the system:

- After deployment is complete, set all SVMs to start automatically with the system. Do not set SVMs under the VMware resource-pool feature.
- In a DirectPath environment, after deploying the system, you must add devices to the SDS.

Other vSphere plug-in features

The plug-in's deployment wizard can be used to register existing PowerFlex systems in vSphere, to add SDS server nodes, or to extend the existing system.

The plug-in can be used to provision PowerFlex nodes.

NOTE: When using DirectPath, also known as PCI-passthrough on the SVM, the following alert appears in vCenter: "Virtual Machine memory Usage". This is a VMWare limitation and can't be resolved. Refer to the VMWare KB for more details:https://kb.vmware.com/s/article/2149787

ESXi Integrated feature support

ESX vStorage APIs for Array Integration (VAAI) is a feature introduced in ESXi/ESX 4.1 that provides hardware acceleration functionality. It allows the host to offload specific virtual machine and storage management operations to compliant storage hardware. With the storage hardware's assistance, the host performs these operations faster, and consumes less CPU, memory, and storage fabric bandwidth.

VAAI uses these fundamental operations:

- Atomic Test & Set (ATS), which is used during creation and locking of files on the VMFS volume
- Clone Blocks/Full Copy/XCOPY, which is used to copy or migrate data within the same physical array
- Zero Blocks/Write Same, which is used to zero-out disk regions
- Thin Provisioning in ESXi 5.x and later hosts, which allows the ESXi host to tell the array when the space previously occupied by a virtual machine (whether it is deleted or migrated to another datastore) can be reclaimed on thin provisioned LUNs.
- Block Delete in ESXi 5.x and later hosts, which allows for space to be reclaimed using the SCSI UNMAP feature.

The PowerFlex supported VAAI features are:

- Atomic Test & Set (ATS)
- Zero Blocks/Write Same
- Thin Provisioning in ESXi 5.x and later hosts
- Block Delete in ESXi 5.x and later hosts

The following output is an example of typical output:

```
esxcli storage core device vaai status get -d
eui.7dbf14034834bbe01bf7e55800000002
eui.7dbf14034834bbe01bf7e55800000002
VAAI Plugin Name:
ATS Status: supported
```

`Clone Status: unsupported` This means that Clone Block/Full Copy/Xcopy is not supported.

`Zero Status: supported` This means that write same is supported.

`Delete Status: supported` This means that UNMAP is supported.

NOTE: Thin provisioning is not shown in VAAI output.

ESXi vStorage APIs for Array Integration (VAAI)

ESXi Integrated feature support.

VMWare SIOC

While Storage I/O Control (SIOC), Storage I/O Statistics Collection (SIOSC) and Network I/O Control (NIOC) are particularly useful for vSan environments, their implementation may actually cause significant issues in a PowerFlex environment, so use of these options is not supported.

PowerFlex provides built-in capabilities to limit network bandwidth and IOPS limits for each volume for each SDC. For more information, refer to "Rebalance throttling", "Rebuild throttling" and "Configuring I/O priorities and bandwidth use" in the *Configure and Customize PowerFlex Guide*.

Prerequisites for adding an NVDIMM to an ESXi 6.7 system

Meet the following prerequisites before adding an NVDIMM to an ESXi 6.7 based PowerFlex system.

Before you begin adding an NVDIMM to an ESXi 6.7 based PowerFlex system, ensure the following:

- You are using VMware version 6.7 for the ESXi host and the vCenter server.
- The VM version of your SVM is version 14 or higher.
- The firmware of the NVDIMM is updated to the version supported by your server. For example, for compatible Dell EMC PowerEdge servers, use NVDIMM firmware version 9324 or higher.
- The ESXi host can detect the NVDIMM.
- You calculate required NVDIMM capacity, according to number of FG devices and their size.
- There is enough NVDIMM capacity on the ESXi host.
- The appropriate VMware licensing is required for persistent memory support; an Enterprise Plus license might be required. To check whether your license supports persistent memory, refer to VMware Compare vSphere Editions and Features.

Verify that the ESXi host can detect the NVDIMM

Before adding an NVDIMM to an ESXi-based PowerFlex system, verify that the ESXi host can detect the NVDIMM.

Prerequisites

Steps

1. Login to the vCenter.
2. Select your ESXi host.
3. Navigate to the **Summary** tab.
4. In the **Hardware** section, verify that the required amount of **Persistent Memory** is listed.

Calculate required NVDIMM and RAM capacity for FG SDS

Use the following formulas to calculate the necessary capacity for NVDIMM and RAM.

About this task

NOTE: When adding NVDIMM capacity to the SVM, the complete capacity of the NVDIMM must be allocated in a single device when adding to the SVM. For example, 16 GB*2 of NVDIMM will be added as 31 GB (rounded down, since VMware uses some of the capacity for datastore management).

Steps

1. Calculate the required NVDIMM and RAM capacity using the following table, where *X* is the total capacity of SDS devices used for the FG storage pool, and *Y* is the total RAM needed:

FG capacity	Required NVDIMM memory size (GiB)	Required RAM capacity (rounded)	Additional memory for MDM, LIA, and SVM OS	Total RAM (rounded)
49 TB	32 GiB NVDIMM (in SVM it will be 31 GiB)	>36 GiB	MDM: 6.5 GiB LIA: 0.35 GiB OS Base: 1 GiB Buffer: 2 GiB CloudLink 4 GiB	50 GiB
49 TB < *X* < 96 TB	64 GiB NVDIMM (in SVM it will be 62 GiB)	>41 GiB <59 GiB		49 GiB < *Y* < 73 GiB
96 TB < *X* < 128 TB (122.88 is actual limit)	96 GiB NVDIMM (in SVM it will be 93 GiB)	>64.5 GiB <72 GiB		86 GiB

2. Alternatively, you can calculate NVDIMM capacity and RAM capacity using the following formulas:

   NOTE: The calculation is in binary MiB, GiB, and TiB

   - *NVDIMM_capacity_in_GiB* = ((100 * *Number_Of_Drives*) + (700 * Capacity_in_TiB))/1024
   - *RAM_capacity_in_GiB* = 10 + ((100 * *Number_Of_Drives*) + (550 * Capacity_in_TiB))/1024

Other functions

This section lists PowerFlex additional functions.

PowerFlex includes the following functions:

- EMC Secure Remote Support (ESRS)

  ESRS support enables secure, high-speed, 24x7, remote connection between EMC and customer installations, including:
  - Remote monitoring
  - Remote diagnosis and repair
  - Daily sending of system events (rsyslog output), alerts, and PowerFlex topology

- Syslog

  The MDM syslog service can send events, via TCP/IP, to RFC 6587-compliant remote (or local) Syslog servers. Messages are sent with facility local0, by default. Once the syslog service is started, all events will be sent until the service is stopped.

- Get Info

  Get Info assembles a ZIP file of system logs for troubleshooting. You can run this function from a local node for its own logs, using the CLI, or by using the PowerFlex Installer to assemble logs from all MDM and SDS nodes in the system. In addition to the log files, a visual snapshot of the PowerFlex GUI, from the time you perform the operation, can be saved, to better enable support options.

  The Get Info function is described in the *Log Collection Guide*.

- Quality of Service (QoS)

  You can adjust the amount of bandwidth and storage that any given SDC can use. You can configure this with the CLI and the REST interface, on a per client/per volume basis.

- Running Remove script on system via Gateway

  In PowerFlex Gateway, you can run a user provided script as part of the Non-disruptive orchestration process (like NDU), which is usually intended for OS patching. This is only supported on Linux, specifically: RHEL and SLES.

  Using this feature includes two main steps:

  1. To manually copy the script file to each PowerFlex host using these script names:
     - patch_script
     - verification_script

2. Copy the script to the `~/lia/bin` folder and add execution permissions.

   RC codes are saved in the LIA log and an error is returned if needed to the Gateway.

3. Execute the scripts from PowerFlex Gateway.

NOTE: You need to test the patch_script and verification_script before running the process on the PowerFlex Gateway

- Background Device Scanner

  The Background Device Scanner ("scanner") enhances the resilience of PowerFlex by constantly searching for, and fixing, device errors before they can affect your system. This provides increased data reliability than the media's checksum scheme provides. The scanner seeks out corrupted sectors of the devices in that pool, provides SNMP reporting about errors found, and keeps statistics about its operation.

  When a scan is completed, the process starts again, thus adding constant protection to your system.

  For Fine Granularity (FG), the Background Device Scanner is enabled by default. It goes through a cycle of each SSD and compares the physical checksums against the data in the Logs and Metadata. The Bandwith Limit is set to 1024 KBs by default.

  You can set the scan rate (default: 1 MB/second per device), which limits the bandwidth allowed for scanning, and choose from the following scan modes:

  - Device only mode

    The scanner uses the device's internal checksum mechanism to validate the primary and secondary data. If a read succeeds in both devices, no action is taken. If a faulty area is read, an error will be generated.

    If a read fails on one device, the scanner attempts to correct the faulty device with the data from the good device. If the fix succeeds, the error-fixes counter is increased. If the fix fails, a device error is issued.

    NOTE:

    A similar algorithm is performed every time an application read fails on the primary device.

    If the read fails on both devices, the scanner skips to the next storage block.

  - Data comparison mode (only available if zero padding is enabled)

    The scanner performs the same algorithm as above, with the following additions:

    After successful reads of the primary and secondary copies of the data, the scanner calculates and compares their checksums. If this comparison fails, the compare errors counter is increased, and the scanner attempts to overwrite the secondary device with the data from the primary device. If this fails, a device error is issued.

    The scanning function is enabled and disabled (default) at the Storage Pool level, and this setting affects all devices in the Storage Pool. You can make these changes at anytime, and you can add/remove volumes and devices while the scanner is enabled.

    When adding a device to a Storage Pool in which the scanner is enabled, the scanning will start about 30 seconds after the device is added.

- AD over LDAP or LDAPS authentication

  User authentication may be done using AD (Active Directory) over LDAP (Lightweight Directory Access Protocol) or LDAPS (Secure LDAP). PowerFlex can support both AD users that are fully controlled through the customer's existing centralized location, and local users (as has been supported in earlier PowerFlex versions). You can associate groups from the AD with the existing PowerFlex roles in order to ensure the Role-Based Access (RBAC) model. When a user logs on to PowerFlex, the MDM identifies that the user belongs to the AD domain, and authenticates the user against the AD server over secured communications. Once the user is authenticated, PowerFlex accepts the group to which the user belongs according to the AD, and associates the appropriate role and its user permissions to that user. The AD implementation is fully redundant.

  NOTE:

  The authorization permissions of each role are defined differently for local authentication, and for LDAP/LDAPS authentication.

  The benefits of using AD over LDAP/LDAPS include:

  - Full control of PowerFlex users through the main user repository
  - No need to specify a local user for each customer

If the AD directory is down, the administrator can always use local users to maintain PowerFlex.

- LDAP for LIA

  You can deploy a system with a LIA user already configured to use LDAP. After upgrade, you can switch LIA from local user to LDAP user. You can add\remove up to eight LDAP servers.

- Multiple LDAP support in the Gateway

  For post deploy, use the FOSGWTool to add LDAP servers to the Gateway login authority. You can add\remove up to eight LDAP servers.

- Oscillating failure handling

  The Oscillating Failures feature detects and reports various oscillating failures, in cases when components fail repeatedly and cause unnecessary failovers, and therefore disruptions to normal system operation. Typical examples of oscillating failures include:
  - A disk that accepts some I/Os and rejects others
  - A node with interrupted connectivity
  - A node that is constantly busy and therefore handles some I/Os too slowly
  - A disk that is sometimes slow to respond
  - A network that is experiencing disruptions

    The smart detection of such failures provides the ability to handle error situations, and to reduce their impact on normal system operation. Oscillating failure handling can be set for MDMs, SDSs and for SDCs. For SDSs, failure handling can be defined per Protection Domain or per Storage Pool.

  - Oscillating failure counters

    The following table describes the oscillating failure counters:

Oscillating failure counters	Description
(sds_sds/sdc_mdm/sdc_sds/mdm_sds) network_disconnections	Measures the number of network disconnections (socket closed) between two components per IP address
sds_decoupled	Measures the number of times an SDS process is down, as detected by the MDM
sds_configuration_failures	Measures the number of times the MDM fails to configure an SDS, when connecting to an SDS (failures occur during the reconfiguration phase)
sds_receive_buffer_allocation_failures	Measures the number of times an SDS fails to allocate buffer for receiving messages
sdc_long_operations	Measures the number of SDC RPC operations that take longer than the predefined threshold (default threshold is 5 seconds)
sdc_memory_allocation_failures	Measures the number of memory allocation failures in each SDC
sdc_socket_allocation_failures	Measures the number of socket allocation failures in each SDC
sds_device_long_successful_ios	Measures the number of successful IOs to an SDS device, which take longer than the predefined threshold (default threshold is 250 milliseconds)

- Secure connectivity with external components

  This feature allows external components to authenticate the MDM. After authentication, communication between the MDM and external components is performed using TLS (Transport Layer Security) protocols.

  Secure communication with the MDM is authenticated by the following components:
  - CLI client

    NOTE:

    If the secure mode is not enabled, modifications are necessary to run SCLI commands.

  - PowerFlex Gateway

- PowerFlex GUI client
- PowerFlex Installer client

  Once added in the trust point, all communications will require authentication, followed by communications over TLS. The same method is employed between the PowerFlex Installer and all LIAs.

Physical layer

This section describes the PowerFlex physical layer and outlines the steps for implementing it.

The physical layer consists of the hardware (servers with storage devices and the network between them) and the PowerFlex software installed on them.

Typically, each SDS is physically located on a separate server.

To implement the physical layer, perform the following steps:

1. Install the MDM component on the MDM nodes in one of the following configurations:
   - Three-node redundant cluster (one primary MDM, one secondary MDM, and one tiebreaker).
   - Five-node redundant cluster (one primary MDM, two secondary MDMs, and two tiebreakers).
   - Single node (one primary MDM).

     NOTE: It is not recommended to use single mode in production systems, except in temporary situations. The MDM contains all the metadata required for system operation. Single Mode has no protection, and exposes the system to a single point of failure.

   MDMs do not require dedicated nodes. They can be installed on nodes hosting other PowerFlex components.
2. Install the SDS component on all nodes that will contribute some, or all, of their physical storage.

   Divide the SDS nodes into Protection Domains. Each SDS can be a member of only one Protection Domain.

   Per Protection Domain, divide the physical storage units into Storage Pools, and optionally, into Fault Sets.
3. Install the SDC component on all nodes on which the application will access the data exposed by the PowerFlex volumes.



Figure 7. Physical layout example—3-node cluster

Communication is done over the existing LAN using standard TCP/IP. The MDM and SDS nodes can be assigned up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

You can perform physical layer setup using the following methods:

- PowerFlex Installer, a web-client based tool
- PowerFlex VMware plug-in a VMware plug-in
- Manual installation procedures

  After completing this installation, the physical layer is ready, and can expose a virtual storage layer.

SAN virtualization layer

This section details the steps for exposing the virtual SAN devices to servers with installed and confiured SDCs.

The MDM cluster manages the entire system. It aggregates the entire storage exposed to it by all the SDSs to generate a virtual layer - virtual SAN storage. Volumes can now be defined over the Storage Pools and can be exposed to the applications as a local storage device using the SDCs.

To expose the virtual SAN devices to your servers (the ones on which you installed and configured SDCs), perform the following:

1. Define volumes. Each volume defined over a Storage Pool is evenly distributed over all members using a RAID protection scheme. By having all SDS members of the Storage Pool participate, PowerFlex ensures:
   - Highest and most stable and consistent performance possible
   - Rapid recovery and redistribution of data
   - Massive IOPS and throughput

   You can define volumes as follows:

   - Thick

     Capacity is allocated immediately, even if not actually used. This can cause capacity to be allocated, but never used, leading to wasted capacity.

     Thick capacity provisioning is limited to available capacity.

   - Thin

     Capacity is "on reserve," but not allocated until actually used. This policy enables more flexibility in provisioning.

     Whereas thick capacity is limited to available capacity, thin capacity provisioning can be oversubscribed, as follows:

     Maximum thin capacity provisioning = 5 * (gross capacity - used capacity)

     When capacity usage reaches the level where it may cause IO errors, alerts are generated. At certain higher capacity levels, volumes (even thin volumes) can no longer be created.

     Example:

     In a system with 3 SDSs, each with 10 TB, there are 30 TB of storage.

     In the system, there is already a thick-provisioned volume that takes up 15 TB of the gross capacity (created by adding a 7.5 TB volume).

     MDM will allow a total of 300 TB gross to be provisioned, and since 15 TB are already allocated, you can add a thin-provisioned volume of 285 TB gross (by adding a 142.5 TB volume) or a thick-provisioned volume of 15 TB gross.

2. Map volumes. Designate which SDCs can access the given volumes. This gives rise to the following:
   - Access control per volume exposed
   - Shared nothing or shared everything volumes

     Once an SDC is mapped to a volume, it immediately gets access to the volume and exposes it locally to the applications as a standard block device. These block devices appear as `/dev/sciniX` where *X* is a letter, starting from "a."

     For example:

     - `/dev/scinia`
     - `/dev/scinib`
   - When a volume is defined on an AIX SDC, one device is created with the following pathnames:
     - A block device, named `/dev/scinidX...n`, where *X* is a number, starting from "0."
     - A raw device, named `/dev/rscinidX...n`, where *X* is a number, starting from "0."

     In general, mapping SDCs to AIX raw devices will yield best performance. If you are using the device to create a filesystem, use the block device.

   - The maximum amount of partitions for the scini disk is 15.
3. The maximum amount of volumes that can be mapped to an SDC is listed in the "Product capabilities and system" table.

   NOTE: SDC mapping is similar to LUN mapping, in the sense that it only allows volume access to clients that were explicitly mapped to the volume.

This is the end of the system setup.

Maintenance

Maintenance of PowerFlex is primarily limited to configuration changes of the physical and virtual layers. It requires minimal user attention. When maintenance or planned restart of an SDS is required, the maintenance mode feature can be used to streamline system operation.

Maintain the physical layer

Adding/removing hardware units and configuring them into PowerFlex can result from scale out, hardware failure, OS patching/upgrade and hardware expansion.

In the physical layer, maintenance is limited to adding and removing hardware units and configuring them into PowerFlex. These operations are usually a result of:

- Scaling out when there is a need for additional capacity. This usually results in adding more storage media to the existing servers, or adding additional servers.
- Hardware failure. In cases where there is a hardware (storage media or server) failure and it needs to be replaced.

  In all of the above cases, the operation will require adding or removing storage capacity from the system. In some cases, it may include adding or removing an entire server, and its associated storage media, from the configuration. As far as PowerFlex is concerned, all of these activities translate to SDS reconfigurations.

  If the removed server is an SDC node, or the server to be added requires exposing storage locally, SDC reconfiguration will happen as well.
- Adding or removing storage media. Add or remove the media from the SDS with which it is associated. PowerFlex will redistribute the data accordingly and seamlessly.
- Adding or removing a node. Add or remove the SDC and SDS residing on the node. PowerFlex will redistribute the data accordingly and seamlessly.
- OS patching \ Upgrade or Hardware expansions. Graceful shutdown\reboot operations with return to fully operation state flow.
- Instant maintenance mode enables you to restart a server that hosts an SDS, with minimal impact on PowerFlex, thus bypassing the disruption and effort caused by disorderly shutdown, Protection Domain shutdown, and orderly shutdown.

Instant maintenance mode (IMM)

Using instant maintenance mode, you can restart a server hosting an SDS without requiring shutdown or interrupting application I/Os. Storing all writes created during maintenance to a dedicated Fault Set prevents data loss in case of a single failure.

Instant maintenance mode enables you to restart a server that hosts an SDS, with minimal impact on PowerFlex, thus bypassing the disruption and effort caused by disorderly shutdown, Protection Domain shutdown, and orderly shutdown.

Whereas PowerFlex always uses two copies of user data, invoking maintenance mode introduces an additional copy that stores all writes created during maintenance to an SDS or Fault Set (created during maintenance) in both a primary location and a new location. This copy prevents data loss if a single failure occurs.

When the SDS or Fault Set is returned from maintenance mode, only the new writes are required to be resynchronized, thus minimizing data transfer during and after the update.

Instant maintenance mode does not interrupt application I/Os; it can be run on any amount of members of a Fault Set; and it can run in parallel on different Protection Domains. While an SDS is in maintenance mode, most PowerFlex operations (like adding a volume) cannot be performed in the Fault Set, Protection Domain, or Storage Pool in which the SDS and its devices reside.

To invoke maintenance mode, the following conditions are required:

- Only one Fault Unit (or standalone SDS) can be in maintenance mode at any given time.
- No other SDSs can be in degraded or failed state (force override can be used).
- There must be adequate space on other SDSs for the additional backup (force override can be used).

> **NOTE:** Use of force override options when entering maintenance mode can lead to data unavailability while maintenance mode is activated.

While an SDS is in maintenance mode, it can be shut down with no danger to data.

Protected maintenance mode (PMM)

PMM is anther type of maintenance mode that ensures no data loss during down time.

PMM is another type of maintenance state where a third copy is created. If there should be a node failure during IMM, then there is no SDS up to serve I/Os for the data. With PMM, a third copy is created before maintenance mode, which ensures that if there is a node failure where the second copy of SDS is required, there is still a full backup of the SDS.

Auto abort PMM

PMM consumes a significant amount of capacity. In some cases, this capacity may be required for more important tasks such as user IO or rebuild.

The purpose of this capability is to abort PMM when approaching the end of capacity, to avoid impacting IO or rebuild. For example, if a single node is entering PMM and a third copy is being created, then a second node fails. Auto abort will make sure that the third copy space is made available to perform a rebuild.

Maintaining the virtualization layer

This section lists the operations that can be performed on volumes exposed by the PowerFlex virtual SAN.

The following operations may be performed on volumes that are exposed by the PowerFlex virtual SAN:

- Add or remove a volume:

  Create or delete a volume in the system.

- Increase volume size:

  Add capacity to a given volume, as needed. The change in volume size occurs seamlessly without interrupting I/O.

- Map and unmap volumes to an SDC:

  This enables or disables access to a volume by an SDC, and thus by an application residing on the same node.

- MDM operations:
  - Move the MDM role to another role
  - Add MDM VIP
  - Upgrade of MDM, SDS, SDC and LIA

Management tools

This section lists the available management clients that can be used with PowerFlex. Many of PowerFlex actions can be performed using more than one tool.

You can provision, maintain, and monitor PowerFlex using the following management clients:

- Command Line Interface (CLI)

  The CLI enables you to perform the entire set of configure, maintain, and monitor activities in PowerFlex.

- Graphical User Interface (GUI)

  The new PowerFlex GUI is HTML-5 based and replaces the Java-based desktop application. PowerFlex GUI provides an enhanced ease of use experience that enables you to perform standard configuration and maintain activities, as well as to monitor the storage system's health and performance. You can use the PowerFlex GUI to view the entire system, and then drill down to different elements. The PowerFlex GUI also known as the PowerFlex presentation server is a web server running on Linux. Refer to "PowerFlex presentation server requirements" for reference to supported web browsers and web server specifications.

- VMware plug-in (plug-in)

  The plug-in enables you to perform basic provision and maintain activities in the VMware environment. In addition, the plug-in provides a wizard to deploy PowerFlex in the VMware environment.

NOTE: Support for Adobe flash player is deprecated as of 2020-12-31. To continue using the PowerFlex vSphere plugin, follow the workaround described in the KB article https://www.dell.com/support/kbdoc/en-us/000181275/scaleio-vxflex-adobe-flash-end-of-life-powerflex-vsphere-plugin-workaround.

- OpenStack

  PowerFlex provides a Cinder driver, which enables interoperation between PowerFlex and an OpenStack cloud operating system.

- CloudIQ

  CloudIQ provides simple storage monitoring and proactive health. It gives you information tailored to your needs, access to near real-time analytics, and the ability to monitor storage systems from anywhere at any time.

- REST gateway

  A REST API can be used to expose monitoring and provisioning via the REST interface. The REST server is installed as part of the PowerFlex Gateway

Many PowerFlex activities can be performed in more than one management tool.

The following tool is also provided:

- PowerFlex Installer.The PowerFlex Installer is used for installing PowerFlex, upgrading, and uninstalling components, as well as running the get-info operation. The PowerFlex Installer is installed as part of the PowerFlex Gateway.

PowerFlex VASA architecture

PowerFlex has implemented VMware's vSphere API for Storage Awareness (VASA). PowerFlex's VASA enables control of PowerFlex storage to be handled by the vCenter administrator. PowerFlex's VASA uses VMware vSphere Virtual Volumes (vVols) to enable VMs hosted on ESXis in the vCenter to have their storage mapped directly to storage in PowerFlex. PowerFlex's implementation uses VASA 2.0 and vVols 1.0. There can be one or three instances of VASA.

The following are the different components of PowerFlex's VASA:

- Virtual Volumes (vVols): A VMware object type that allows the vSphere Admin to provision VMs without depending on the Storage Admin.
- vVol datastores: A type of VMware datastore, in addition to VMFS and NFS datastores, which allow vVols to map directly to the storage system. vVol datastores are more granular than VMFS and NFS datastores, enabling VMs or virtual disks to be managed independently. You can create vVol datastores based on one or more underlying storage pools and then allocate the pool to be used for the vVol datastore and its associated vVols.
- Storage containers: A logical abstraction on to which vVols are mapped and stored. In PowerFlex's VASA, the storage container is mapped to a PowerFlex Storage Pool. vSphere maps storage containers to VVol datastores and provides applicable datastore level functionality.
- Storage policies: Storage policies are used to ensure that VMs are placed on storage that guarantees a specific level of performance.
- VASA Storage Virtual Machine (SVM): The virtual machine that runs the VASA.
- VASA-database: A database created to store the VVols metadata.

VASA allows for automatic mapping of storage. When a new VM is created, a request for storage passes through the VASA. The VASA sends the request to the MDM to create the volume. The MDM creates the volume and passes the volume ID back through the VASA to the ESXi server. Mapping volume requests also pass through the VASA as the MDM maps the volume the VM runs on to the ESXi. For every VM that is created, a minimum of two vVols are created. One is for the VM metadata and one is for the VM disk. When the VM is powered on, another vVol, called the swap vVol, is created. The swap vVol is removed when the VM is powered off. The VASA is used only for storage management; no I/Os are sent through the VASA.

PowerFlex VASA provider limitations and prerequisites

Note the following limitations when using PowerFlex VASA provider. For additional details on the VASA provider server requirements, see the Getting to Know Dell PowerFlex Guide.

Limitations

- The VASA provider can be mapped to a single instance of PowerFlex.
- A cluster may contain a single VASA provider or three VASA providers.
- The VASA provider must be installed only on a node which contains an SDC. The VASA provider nodes can be standalone or shared with other PowerFlex components.
- The VASA VM can be hosted on the same node with SDS nodes. The PowerFlex Gateway and MDM hosts can't host a VASA VM.
- The VASA provider should be registered with a single vCenter.
- PowerFlex limits the maximum snapshot hierarchy of a vVol to 31.
- PowerFlex limits the maximum number of volumes to 130K (8,192 mapped volumes per ESXi).

Configuring direct attached storage (DAS)

When adding DAS to SDS, it is recommended to configure the raw devices as stand-alone devices. Upon adding a device to an SDS PowerFlex verifies that the device is clear and returns an error if it is otherwise.

PowerFlex works with any free capacity—internal or direct-attached devices, either magnetic hard disk drives (HDD) or flash-based devices such as solid state drive (SSD) and PCIe cards. Although PowerFlex can work with any device topology, it is recommended to configure the raw devices as stand-alone devices.

Device data is erased when devices are added to SDS. When adding a device to an SDS, PowerFlex will check that the device is clear before adding it. An error will be returned, per device, if it is found not to be clear. You can override this check by using the force device takeover option.

The following devices are considered to be not "clear," and thus cannot be added to SDS:

- Linux - A complete device with either a filesystem or partition, or a partitioned device with a filesystem.
  - Multipath devices can't be added as SDS devices
  - Devices in an LVM group can't be added to an SDS
- ESXi - Same as above, depending on the OS of the SVM where the SDS is installed.

Limitations:

- SAN devices will not be prevented from being added.
- Devices in an LVM group cannot be added to an SDS.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

NOTE:

If the server has a RAID controller, PowerFlex prefers to use the controller's caching abilities for better performance, but is better utilized when all devices are configured as stand-alone (i.e. setting each of the devices to RAID-0 separately). For HDD devices, it is recommended to enable RAID-controller caching. As for SSD devices, caching is not needed.

NOTE:

For HDDs: It is recommended to use RAID-controller caching when available as follows:

- READ/WRITE: if cache is battery-backed
- READ ONLY: if cache is NOT battery-backed

For SSD's (e.g. Flash): setting should be write-through